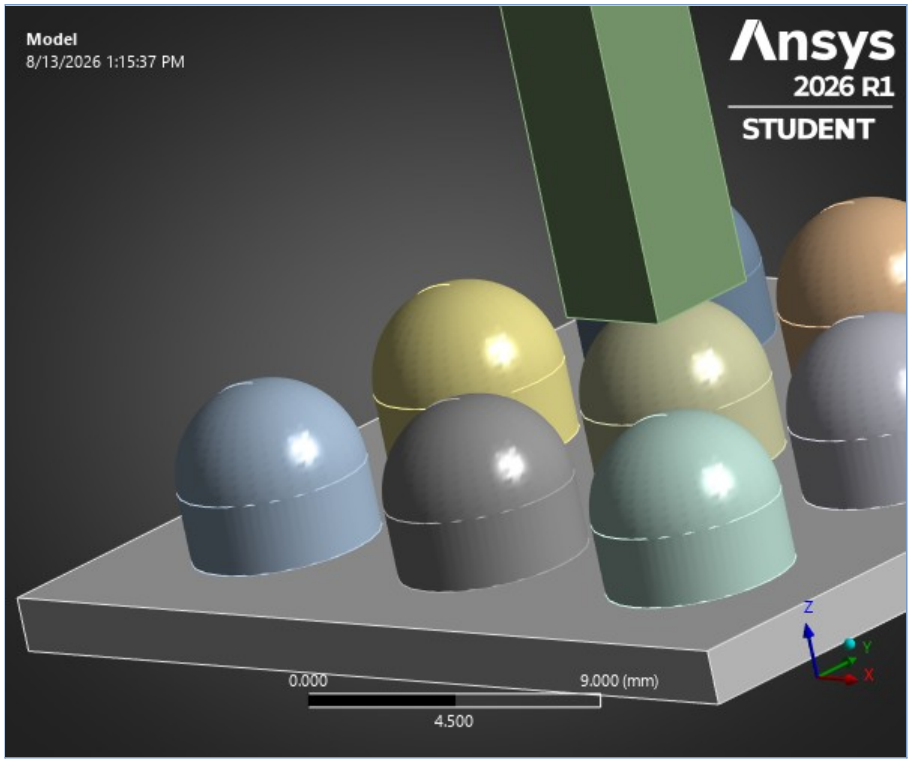
# Project*

| | |
|---|---|
| First Saved | Tuesday, August 11, 2026 |
| Last Saved | Tuesday, August 11, 2026 |
| Product Version | 2026 R1 |
| Save Project Before Solution | No |
| Save Project After Solution | No |



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- **Material Data**
  - ecoflex 00-45
  - Plastic, ABS (high-impact)


# Units

**TABLE 1**

| | |
|---|---|
| Unit System | Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius |
| Angle | Degrees |
| Rotational Velocity | rad/s |
| Temperature | Celsius |

# Model (C4)

**TABLE 2**
**Model (C4) > Geometry Imports**

| | |
|---|---|
| Object Name | *Geometry Imports* |
| State | Solved |

**TABLE 3**
**Model (C4) > Geometry Imports > Geometry Import (C3)**

| | |
|---|---|
| Object Name | *Geometry Import (C3)* |
| State | Solved |
| **Definition** | |
| Source | C:\Users\DLSU\Downloads\RUN 6 (6MM).step |
| Type | Step |
| **Basic Geometry Options** | |
| Solid Bodies | Yes |
| Surface Bodies | Yes |
| Line Bodies | No |
| Parameters | Independent |
| Parameter Key | ANS;DS |
| Attributes | No |
| Named Selections | No |
| Material Properties | No |
| **Advanced Geometry Options** | |
| Use Associativity | Yes |
| Coordinate Systems | No |
| Reader Mode Saves Updated File | No |
| Use Instances | Yes |
| Smart CAD Update | Yes |
| Compare Parts On Update | No |

| | |
|---|---|
| Analysis Type | 3-D |
| Mixed Import Resolution | None |
| Import Facet Quality | Source |
| Clean Bodies On Import | No |
| Stitch Surfaces On Import | None |
| Decompose Disjoint Geometry | Yes |
| Enclosure and Symmetry Processing | Yes |

## *Geometry*

**TABLE 4**
**Model (C4) > Geometry**

| | |
|---|---|
| Object Name | *Geometry* |
| State | Fully Defined |
| **Definition** | |
| Source | C:\Users\DLSU\Downloads\RUN 6 (6MM).step |
| Type | Step |
| Length Unit | Millimeters |
| Element Control | Program Controlled |
| Display Style | Body Color |
| **Bounding Box** | |
| Length X | 30. mm |
| Length Y | 30. mm |
| Length Z | 45.05 mm |
| **Properties** | |
| Volume | 3019.3 mm³ |
| Mass | 3.1822e-003 kg |
| Scale Factor Value | 1. |
| **Statistics** | |
| Bodies | 11 |
| Active Bodies | 11 |
| Nodes | 7174 |
| Elements | 2769 |
| Mesh Metric | None |
| **Update Options** | |
| Assign Default Material | No |
| **Basic Geometry Options** | |
| Solid Bodies | Yes |
| Surface Bodies | Yes |
| Line Bodies | No |
| Parameters | Independent |
| Parameter Key | ANS;DS |
| Attributes | No |
| Named Selections | No |
| Material Properties | No |
| **Advanced Geometry Options** | |
| Use Associativity | Yes |
| Coordinate Systems | No |
| Reader Mode Saves Updated File | No |
| Use Instances | Yes |
| Smart CAD Update | Yes |
| Compare Parts On Update | No |
| Analysis Type | 3-D |
| Mixed Import Resolution | None |
| Import Facet Quality | Source |
| Clean Bodies On Import | No |
| Stitch Surfaces On Import | None |
| Decompose Disjoint Geometry | Yes |
| ID_GeometryPrefProcessPhysicsDefinition | No |
| Enclosure and Symmetry Processing | Yes |

**TABLE 5**
**Model (C4) > Geometry > Parts**

| Object Name | *RUN 6 (6MM) \|Body2* | *RUN 6 (6MM) \|Body3* | *RUN 6 (6MM) \|Body4* | *RUN 6 (6MM) \|Body5* | *RUN 6 (6MM) \|Body6* | *RUN 6 (6MM) \|Body7* | *RUN 6 (6MM) \|Body8* | *RUN 6 (6MM) \|Body9* | *RUN 6 (6MM) \|Body10* | *RUN 6 (6MM) \|Body11* | *RUN 6 (6MM) \|Body12* |
|---|---|---|---|---|---|---|---|---|---|---|---|
| State | Meshed | | | | | | | | | | |
| **Graphics Properties** | | | | | | | | | | | |
| Visible | Yes | | | | | | | | | | |
| Transparency | 1 | | | | | | | | | | |

| Definition | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Suppressed | No | | | | | | | | | | |
| Stiffness Behavior | Flexible | | | | | | | | | | |
| Coordinate System | Default Coordinate System | | | | | | | | | | |
| Reference Temperature | By Environment | | | | | | | | | | |
| Treatment | None | | | | | | | | | | |
| **Material** | | | | | | | | | | | |
| Assignment | ecoflex 00-45 | | | | | | | | | | Plastic, ABS (high-impact) |
| Nonlinear Effects | Yes | | | | | | | | | | |
| Thermal Strain Effects | Yes | | | | | | | | | | |
| **Bounding Box** | | | | | | | | | | | |
| Length X | 30. mm | 6. mm | | | | | | | | | 4. mm |
| Length Y | 30. mm | 6. mm | | | | | | | | | 4. mm |
| Length Z | 1.75 mm | 5.25 mm | | | | | | | | | 38. mm |
| **Properties** | | | | | | | | | | | |
| Volume | 1498.2 mm³ | 101.46 mm³ | | | | | | | | | 608. mm³ |
| Mass | 1.5881e-003 kg | 1.0755e-004 kg | | | | | | | | | 6.2624e-004 kg |
| Centroid X | -3.5195e-016 mm | -9. mm | 9. mm | -1.1112e-009 mm | 9. mm | -9. mm | 9. mm | -3.1362e-010 mm | 5.9581e-010 mm | -9. mm | 0. mm |
| Centroid Y | -9.6632e-017 mm | -9. mm | 9. mm | | -9. mm | -6.4891e-010 mm | 5.7823e-010 mm | -1.116e-010 mm | -9. mm | 9. mm | 7.791e-018 mm |
| Centroid Z | -0.875 mm | 2.2354 mm | | | | | | | | | 24.3 mm |
| Moment of Inertia Ip1 | 0.12119 kg·mm² | 4.469e-004 kg·mm² | | | | | | | | | 7.6193e-002 kg·mm² |
| Moment of Inertia Ip2 | 0.12119 kg·mm² | 4.469e-004 kg·mm² | | | | | | | | | 7.6193e-002 kg·mm² |
| Moment of Inertia Ip3 | 0.24157 kg·mm² | 5.0321e-004 kg·mm² | | | | | | | | | 1.67e-003 kg·mm² |
| **Statistics** | | | | | | | | | | | |
| Nodes | 1657 | 533 | 534 | 549 | 541 | 598 | 571 | 601 | 528 | 561 | 501 |
| Elements | 198 | 260 | 259 | 274 | 264 | 308 | 291 | 313 | 259 | 279 | 64 |
| Mesh Metric | None | | | | | | | | | | |

**TABLE 6**
**Model (C4) > Materials**

| Object Name | *Materials* |
|---|---|
| State | Fully Defined |
| **Statistics** | |
| Materials | 3 |
| Material Assignments | 0 |

## *Coordinate Systems*

**TABLE 7**
**Model (C4) > Coordinate Systems > Coordinate System**

| Object Name | *Global Coordinate System* |
|---|---|
| State | Fully Defined |
| **Definition** | |
| Type | Cartesian |
| Coordinate System ID | 0. |
| **Origin** | |
| Origin X | 0. mm |
| Origin Y | 0. mm |
| Origin Z | 0. mm |
| **Directional Vectors** | |
| X Axis Data | [ 1. 0. 0. ] |
| Y Axis Data | [ 0. 1. 0. ] |
| Z Axis Data | [ 0. 0. 1. ] |
| **Transfer Properties** | |
| Source | |
| Read Only | No |

## Connections

**TABLE 8**
**Model (C4) > Connections**

| Object Name | *Connections* |
|---|---|
| State | Fully Defined |
| **Auto Detection** | |
| Generate Automatic Connection On Refresh | Yes |
| **Transparency** | |
| Enabled | Yes |
| **Statistics** | |
| Contacts | 10 |
| Active Contacts | 10 |
| Joints | 0 |
| Active Joints | 0 |
| Beams | 0 |
| Active Beams | 0 |
| Bearings | 0 |
| Active Bearings | 0 |
| Springs | 0 |
| Active Springs | 0 |
| Body Interactions | 0 |
| Active Body Interactions | 0 |

**TABLE 9**
**Model (C4) > Connections > Contacts**

| Object Name | *Contacts* | *Contacts 2* |
|---|---|---|
| State | Fully Defined | |
| **Definition** | | |
| Connection Type | Contact | |
| **Scope** | | |
| Scoping Method | Geometry Selection | |
| Geometry | All Bodies | |
| **Auto Detection** | | |
| Tolerance Type | Slider | |
| Tolerance Slider | 0. | |
| Tolerance Value | 0.15471 mm | |
| Use Range | No | |
| Face/Face | Yes | |
| Face-Face Angle Tolerance | 75. ° | |
| Face Overlap Tolerance | Off | |
| Cylindrical Faces | Include | |
| Face/Edge | No | |
| Edge/Edge | No | |
| Priority | Include All | |
| Group By | Bodies | |
| Search Across | Bodies | |
| **Statistics** | | |
| Connections | 9 | 1 |
| Active Connections | 9 | 1 |

**TABLE 10**
**Model (C4) > Connections > Contacts > Contact Regions**

| Object Name | *Contact Region* | *Contact Region 2* | *Contact Region 3* | *Contact Region 4* | *Contact Region 5* | *Contact Region 6* | *Contact Region 7* | *Contact Region 8* | *Contact Region 9* |
|---|---|---|---|---|---|---|---|---|---|
| State | Fully Defined | | | | | | | | |
| **Scope** | | | | | | | | | |
| Scoping Method | Geometry Selection | | | | | | | | |
| Contact | 1 Face | | | | | | | | |
| Target | 1 Face | | | | | | | | |
| Contact Bodies | RUN 6 (6MM)\|Body2 | | | | | | | | |
| Target Bodies | RUN 6 (6MM) \|Body3 | RUN 6 (6MM) \|Body4 | RUN 6 (6MM) \|Body5 | RUN 6 (6MM) \|Body6 | RUN 6 (6MM) \|Body7 | RUN 6 (6MM) \|Body8 | RUN 6 (6MM) \|Body9 | RUN 6 (6MM) \|Body10 | RUN 6 (6MM) \|Body11 |
| Protected | No | | | | | | | | |
| **Definition** | | | | | | | | | |
| Type | Bonded | | | | | | | | |
| Scope Mode | Automatic | | | | | | | | |
| Behavior | Program Controlled | | | | | | | | |
| Trim Contact | Program Controlled | | | | | | | | |
| Trim Tolerance | 0.15471 mm | | | | | | | | |

| | |
|---|---|
| Contact APDL Name | |
| Target APDL Name | |
| Suppressed | No |
| **Display** | |
| Element Normals | No |
| **Advanced** | |
| Formulation | Program Controlled |
| Small Sliding | Program Controlled |
| Detection Method | Program Controlled |
| Penetration Tolerance | Program Controlled |
| Elastic Slip Tolerance | Program Controlled |
| Normal Stiffness | Program Controlled |
| Update Stiffness | Program Controlled |
| Pinball Region | Program Controlled |
| **Geometric Modification** | |
| Contact Geometry Correction | None |
| Target Geometry Correction | None |

**TABLE 11**
**Model (C4) > Connections > Contacts 2 > Contact Regions**

| | |
|---|---|
| Object Name | *Frictionless - RUN 6 (6MM)\|Body9 To RUN 6 (6MM)\|Body12* |
| State | Fully Defined |
| **Scope** | |
| Scoping Method | Geometry Selection |
| Contact | 1 Face |
| Target | 1 Face |
| Contact Bodies | RUN 6 (6MM)\|Body9 |
| Target Bodies | RUN 6 (6MM)\|Body12 |
| Protected | No |
| **Definition** | |
| Type | Frictionless |
| Scope Mode | Manual |
| Behavior | Asymmetric |
| Trim Contact | Program Controlled |
| Contact APDL Name | |
| Target APDL Name | |
| Suppressed | No |
| **Display** | |
| Element Normals | No |
| **Advanced** | |
| Formulation | Augmented Lagrange |
| Small Sliding | Program Controlled |
| Detection Method | Program Controlled |
| Penetration Tolerance | Program Controlled |
| Normal Stiffness | Program Controlled |
| Update Stiffness | Program Controlled |
| Stabilization Damping Factor | 0. |
| Pinball Region | Program Controlled |
| Time Step Controls | None |
| **Geometric Modification** | |
| Interface Treatment | Add Offset, No Ramping |
| Offset | 0. mm |
| Contact Geometry Correction | None |
| Target Geometry Correction | None |

## *Mesh*

**TABLE 12**
**Model (C4) > Mesh**

| | |
|---|---|
| Object Name | *Mesh* |
| State | Solved |
| **Display** | |
| Display Style | Use Geometry Setting |
| **Defaults** | |
| Physics Preference | Mechanical |

| | |
|---|---|
| Element Order | Program Controlled |
| Element Size | Default |
| **Sizing** | |
| Mesh Sizing | Adaptive |
| Resolution | Default (2) |
| Mesh Defeaturing | Yes |
| Defeature Size | Default |
| Transition | Fast |
| Span Angle Center | Coarse |
| Initial Size Seed | Assembly |
| Bounding Box Diagonal | 61.883 mm |
| Average Surface Area | 62.105 mm² |
| Minimum Edge Length | 1.75 mm |
| **Quality** | |
| Check Mesh Quality | Yes, Errors |
| Error Limits | Aggressive Mechanical |
| Target Element Quality | Default (0.1) |
| Smoothing | Medium |
| Mesh Metric | None |
| **Inflation** | |
| Use Automatic Inflation | None |
| Inflation Option | Smooth Transition |
| Transition Ratio | 0.272 |
| Maximum Layers | 5 |
| Growth Rate | 1.2 |
| Inflation Algorithm | Pre |
| Inflation Element Type | Wedges |
| View Advanced Options | No |
| **Advanced** | |
| Number of CPUs for Parallel Part Meshing | Program Controlled |
| Straight Sided Elements | No |
| Rigid Body Behavior | Dimensionally Reduced |
| Triangle Surface Mesher | Program Controlled |
| Topology Checking | Yes |
| Pinch Tolerance | Please Define |
| Generate Pinch on Refresh | No |
| Auto-Map Fillets | No |
| **Automatic Methods** | |
| Sheet Body Method | Prime Quad Dominant |
| Sweepable Body Method | Sweep |
| **Statistics** | |
| Nodes | 7174 |
| Elements | 2769 |
| Show Detailed Statistics | No |

## Static Structural (C5)

**TABLE 13**
**Model (C4) > Analysis**

| | |
|---|---|
| Object Name | *Static Structural (C5)* |
| State | Solved |
| **Definition** | |
| Physics Type | Structural |
| Analysis Type | Static Structural |
| Solver Target | Mechanical APDL |
| **Options** | |
| Environment Temperature | 22. °C |
| Generate Input Only | No |

**TABLE 14**
**Model (C4) > Static Structural (C5) > Analysis Settings**

| | |
|---|---|
| Object Name | *Analysis Settings* |
| State | Fully Defined |
| **Step Controls** | |
| Number Of Steps | 1. |
| Current Step Number | 1. |
| Step End Time | 1. s |
| Auto Time Stepping | On |
| Define by | Substeps |
| Initial Substeps | 20. |

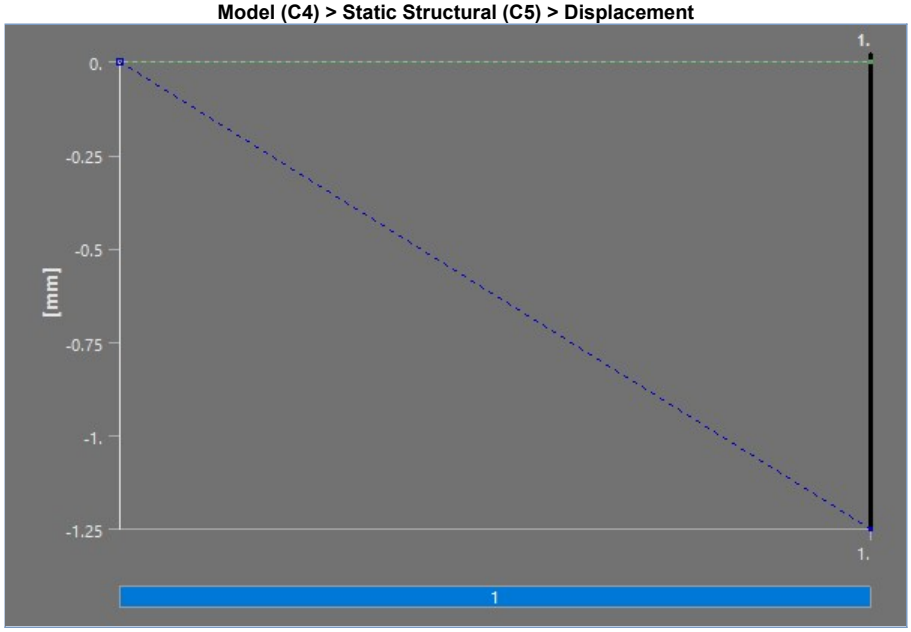
| | |
|---|---|
| Minimum Substeps | 5. |
| Maximum Substeps | 500. |
| Maximum Iterations Per Substep | Program Controlled |
| **Solver Controls** | |
| Solver Type | Program Controlled |
| Weak Springs | On |
| Spring Stiffness | Program Controlled |
| Solver Pivot Checking | Program Controlled |
| Large Deflection | On |
| Inertia Relief | Off |
| Quasi-Static Solution | Off |
| **Rotordynamics Controls** | |
| Coriolis Effect | Off |
| **Restart Controls** | |
| Generate Restart Points | Program Controlled |
| Retain Files After Full Solve | No |
| Combine Restart Files | Program Controlled |
| **Nonlinear Controls** | |
| Newton-Raphson Option | Program Controlled |
| Force Convergence | Program Controlled |
| Moment Convergence | Program Controlled |
| Displacement Convergence | Program Controlled |
| Rotation Convergence | Program Controlled |
| Line Search | Program Controlled |
| Stabilization | Program Controlled |
| **Advanced** | |
| Inverse Option | No |
| Contact Split (DMP) | Program Controlled |
| **Output Controls** | |
| Output Selection | None |
| Stress | Yes |
| Back Stress | No |
| Strain | Yes |
| Contact Data | Yes |
| Nonlinear Data | No |
| Nodal Forces | No |
| Volume and Energy | Yes |
| Euler Angles | Yes |
| General Miscellaneous | No |
| Contact Miscellaneous | No |
| Store Results At | All Time Points |
| **Result File Options** | |
| Result File Compression | Program Controlled |
| **Analysis Data Management** | |
| Solver Files Directory | C:\Users\DLSU\Downloads\THESIS DATA\ANSYS FOR HOLLOW RODS\solid v hollow and diameter_files\dp0\SYS-2\MECH\ |
| Future Analysis | None |
| Scratch Solver Files Directory | |
| Save MAPDL db | No |
| Contact Summary | Program Controlled |
| Delete Unneeded Files | Yes |
| Nonlinear Solution | Yes |
| Solver Units | Active System |
| Solver Unit System | nmm |

**TABLE 15**
**Model (C4) > Static Structural (C5) > Loads**

| Object Name | *Fixed Support* | *Displacement* |
|---|---|---|
| State | Fully Defined | |
| **Scope** | | |
| Scoping Method | Geometry Selection | |
| Geometry | 1 Face | |
| **Definition** | | |
| Type | Fixed Support | Displacement |
| Suppressed | No | |
| Define by | | Components |
| Coordinate System | | Global Coordinate System |
| X Component | | 0. mm (ramped) |
| Y Component | | 0. mm (ramped) |
| | | |

| | | |
|---|---|---|
| Z Component | | -1.25 mm (ramped) |

**FIGURE 1**
**Model (C4) > Static Structural (C5) > Displacement**




## *Solution (C6)*

**TABLE 16**
**Model (C4) > Static Structural (C5) > Solution**

| Object Name | *Solution (C6)* |
|---|---|
| State | Solved |
| **Adaptive Mesh Refinement** | |
| Max Refinement Loops | 1. |
| Refinement Depth | 2. |
| **Information** | |
| Status | Done |
| MAPDL Elapsed Time | 48.825 s |
| MAPDL Memory Used | 241. MB |
| MAPDL Result File Size | 14.188 MB |
| **Post Processing** | |
| Beam Section Results | No |
| On Demand Stress/Strain | No |

**TABLE 17**
**Model (C4) > Static Structural (C5) > Solution (C6) > Solution Information**

| Object Name | *Solution Information* |
|---|---|
| State | Solved |
| **Solution Information** | |
| Solution Output | Solver Output |
| Newton-Raphson Residuals | 0 |
| Identify Element Violations | 0 |
| Update Interval | 2.5 s |
| Display Points | All |
| **FE Connection Visibility** | |
| Activate Visibility | Yes |
| Display | All FE Connectors |
| Draw Connections Attached To | All Nodes |
| Line Color | Connection Type |
| Visible on Results | No |
| Line Thickness | Single |
| Display Type | Lines |

**TABLE 18**
**Model (C4) > Static Structural (C5) > Solution (C6) > Results**

| Object Name | *Total Deformation* | *Directional Deformation* | *Directional Deformation 2* | *Directional Deformation 3* | *Equivalent Elastic Strain* | *Maximum Principal Elastic* | *Maximum Shear Elastic* | *Elastic Strain Intensity* | *Equivalent Stress* | *Maximum Principal Stress* | *Maximum Shear Stress* |
|---|---|---|---|---|---|---|---|---|---|---|---|

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Strain* | *Strain* | | | | |
| State | Solved | | | | | | | | | | |
| **Scope** | | | | | | | | | | | |
| Scoping Method | Geometry Selection | | | | | | | | | | |
| Geometry | 1 Body | | | | | | | | | | |
| **Definition** | | | | | | | | | | | |
| Type | Total Deformation | Directional Deformation | | | Equivalent Elastic Strain | Maximum Principal Elastic Strain | Maximum Shear Elastic Strain | Elastic Strain Intensity | Equivalent (von-Mises) Stress | Maximum Principal Stress | Maximum Shear Stress |
| By | Time | | | | | | | | | | |
| Display Time | Last | | | | | | | | | | |
| Separate Data by Entity | No | | | | | | | | | | |
| Calculate Time History | Yes | | | | | | | | | | |
| Identifier | | | | | | | | | | | |
| Suppressed | No | | | | | | | | | | |
| Orientation | | X Axis | Y Axis | Z Axis | | | | | | | |
| Coordinate System | | Global Coordinate System | | | | | | | | | |
| **Results** | | | | | | | | | | | |
| Minimum | 2.3789e-002 mm | -0.26077 mm | -0.23785 mm | -1.1931 mm | 5.0826e-002 mm/mm | 2.8335e-002 mm/mm | 7.7796e-002 mm/mm | | 6.7548e-003 MPa | -2.6782e-002 MPa | 3.6168e-003 MPa |
| Maximum | 1.1939 mm | 0.2358 mm | 0.23178 mm | -2.1564e-002 mm | 0.28769 mm/mm | 0.16847 mm/mm | 0.44251 mm/mm | | 3.5862e-002 MPa | 1.389e-002 MPa | 1.9719e-002 MPa |
| Average | 0.32926 mm | -4.11e-003 mm | -3.7474e-003 mm | -0.28609 mm | 0.12745 mm/mm | 7.5989e-002 mm/mm | 0.18891 mm/mm | | 1.5816e-002 MPa | 1.1424e-003 MPa | 8.6032e-003 MPa |
| Minimum Occurs On | RUN 6 (6MM)\|Body9 | | | | | | | | | | |
| Maximum Occurs On | RUN 6 (6MM)\|Body9 | | | | | | | | | | |
| **Minimum Value Over Time** | | | | | | | | | | | |
| Minimum | 8.9294e-017 mm | -0.26077 mm | -0.23785 mm | -1.1931 mm | 6.8075e-016 mm/mm | -9.7026e-017 mm/mm | 1.2136e-017 mm/mm | | 1.9063e-017 MPa | -2.6782e-002 MPa | 1.0212e-017 MPa |
| Maximum | 2.3789e-002 mm | -2.994e-015 mm | -9.77e-015 mm | -5.6811e-015 mm | 5.0826e-002 mm/mm | 2.8335e-002 mm/mm | 7.7796e-002 mm/mm | | 6.7548e-003 MPa | -4.2383e-015 MPa | 3.6168e-003 MPa |
| **Maximum Value Over Time** | | | | | | | | | | | |
| Minimum | 1.0734e-014 mm | 4.1078e-015 mm | 6.6613e-015 mm | -2.1564e-002 mm | 5.4725e-015 mm/mm | 2.8866e-015 mm/mm | 6.9019e-015 mm/mm | | 9.4577e-016 MPa | 7.0866e-015 MPa | 5.4528e-016 MPa |
| Maximum | 1.1939 mm | 0.2358 mm | 0.23178 mm | 3.2385e-015 mm | 0.28769 mm/mm | 0.16847 mm/mm | 0.44251 mm/mm | | 3.5862e-002 MPa | 1.389e-002 MPa | 1.9719e-002 MPa |
| **Information** | | | | | | | | | | | |
| Time | 1. s | | | | | | | | | | |
| Load Step | 1 | | | | | | | | | | |
| Substep | 9 | | | | | | | | | | |
| Iteration Number | 30 | | | | | | | | | | |
| **Integration Point Results** | | | | | | | | | | | |
| Display Option | | | | | Averaged | | | | | | |
| Average Across Bodies | | | | | No | | | | | | |

**FIGURE 2**
**Model (C4) > Static Structural (C5) > Solution (C6) > Total Deformation**

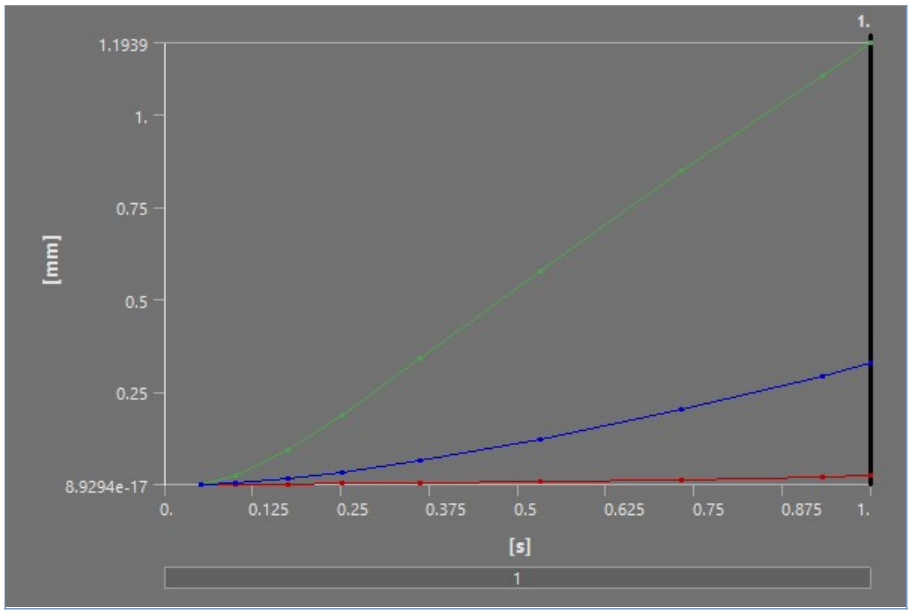



**TABLE 19**
**Model (C4) > Static Structural (C5) > Solution (C6) > Total Deformation**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | 8.9294e-017 | 1.0734e-014 | 1.5469e-015 |
| 0.1 | 2.8301e-004 | 2.4538e-002 | 3.8033e-003 |
| 0.175 | 1.3249e-003 | 9.3217e-002 | 1.5158e-002 |
| 0.25 | 2.6809e-003 | 0.18569 | 3.2224e-002 |
| 0.3625 | 4.9479e-003 | 0.34032 | 6.4486e-002 |
| 0.53125 | 8.7767e-003 | 0.57846 | 0.12278 |
| 0.73125 | 1.3985e-002 | 0.84832 | 0.20179 |
| 0.93125 | 2.1124e-002 | 1.1061 | 0.29324 |
| 1. | 2.3789e-002 | 1.1939 | 0.32926 |

**FIGURE 3**
**Model (C4) > Static Structural (C5) > Solution (C6) > Total Deformation > Figure**

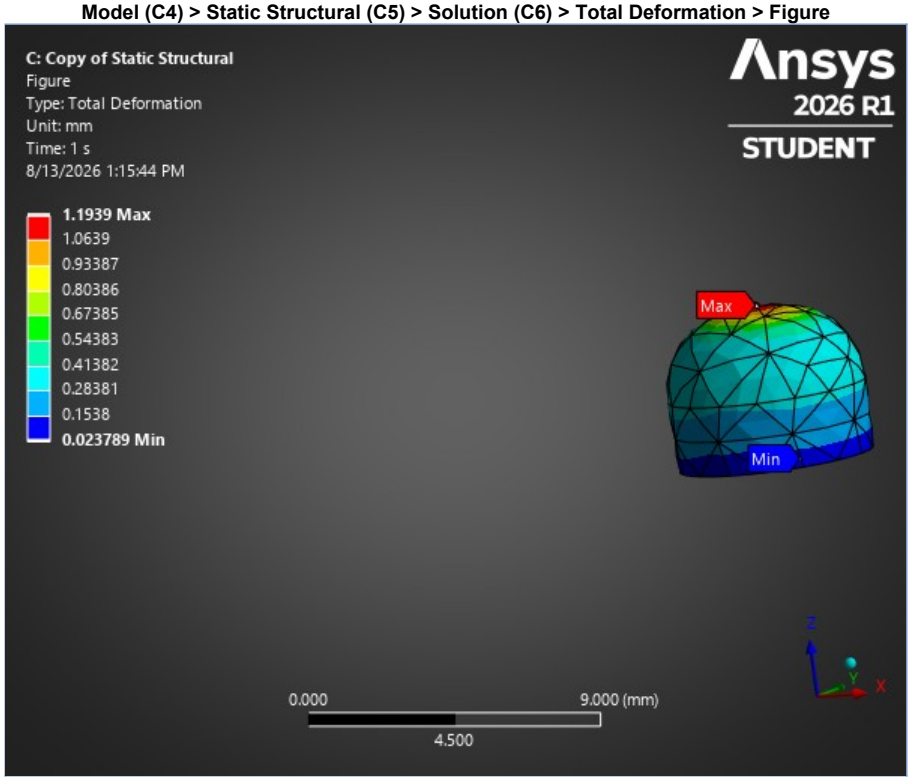



**FIGURE 4**

**Model (C4) > Static Structural (C5) > Solution (C6) > Directional Deformation**

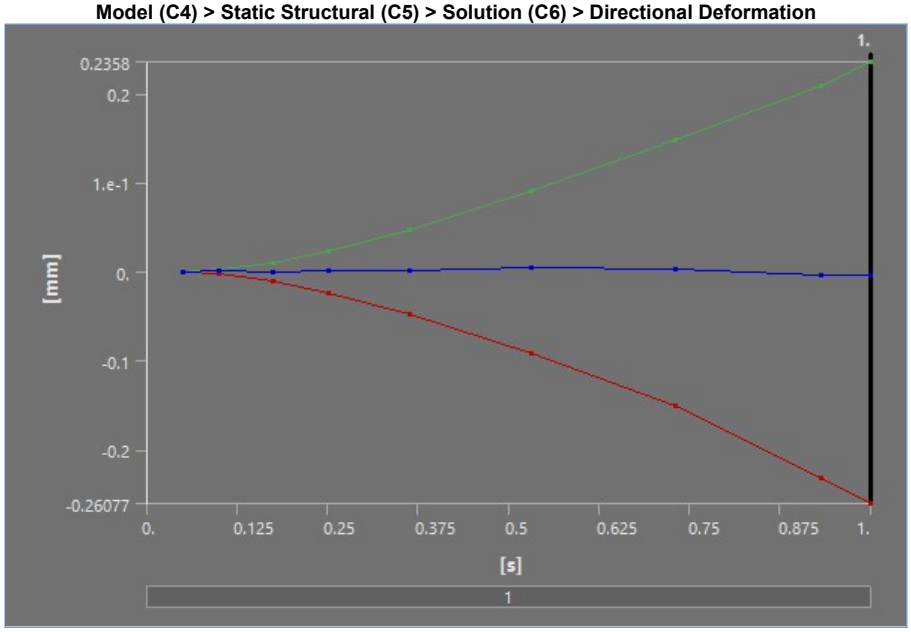



**TABLE 20**
**Model (C4) > Static Structural (C5) > Solution (C6) > Directional Deformation**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | -2.994e-015 | 4.1078e-015 | -7.5577e-016 |
| 0.1 | -2.496e-003 | 2.9674e-003 | 2.9114e-004 |
| 0.175 | -1.1635e-002 | 1.0205e-002 | -2.2465e-006 |
| 0.25 | -2.4042e-002 | 2.2652e-002 | 6.6224e-004 |
| 0.3625 | -4.8701e-002 | 4.6408e-002 | 1.8312e-003 |
| 0.53125 | -9.2321e-002 | 9.0097e-002 | 3.6176e-003 |
| 0.73125 | -0.15181 | 0.14774 | 3.261e-003 |
| 0.93125 | -0.23189 | 0.2096 | -3.4599e-003 |
| 1. | -0.26077 | 0.2358 | -4.11e-003 |

**FIGURE 5**
**Model (C4) > Static Structural (C5) > Solution (C6) > Directional Deformation > Figure**

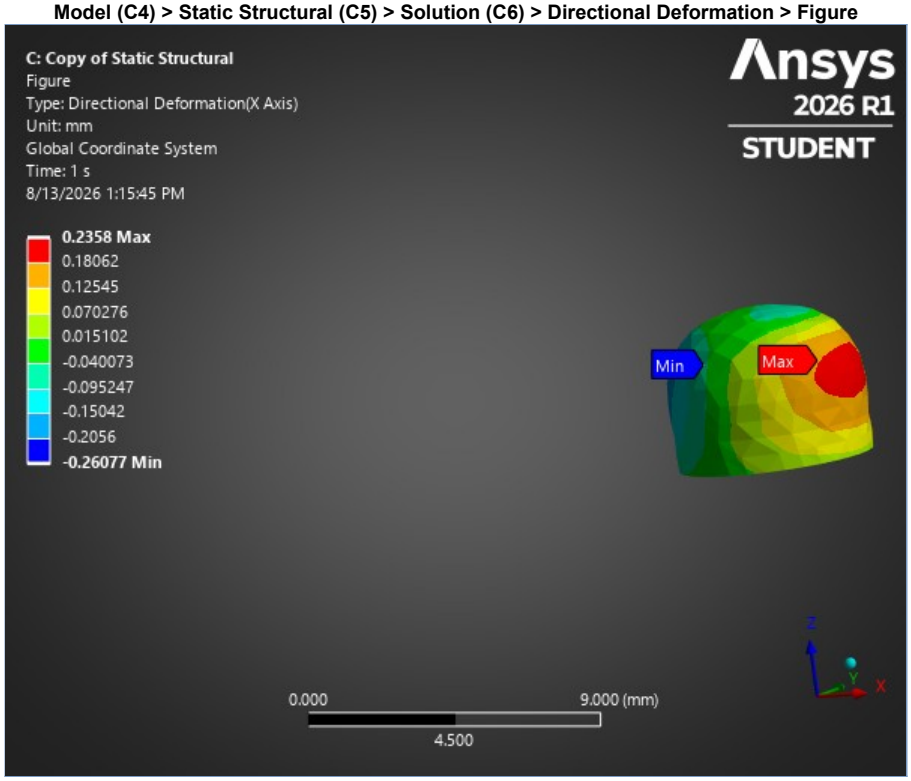


**FIGURE 6**
**Model (C4) > Static Structural (C5) > Solution (C6) > Directional Deformation 2**

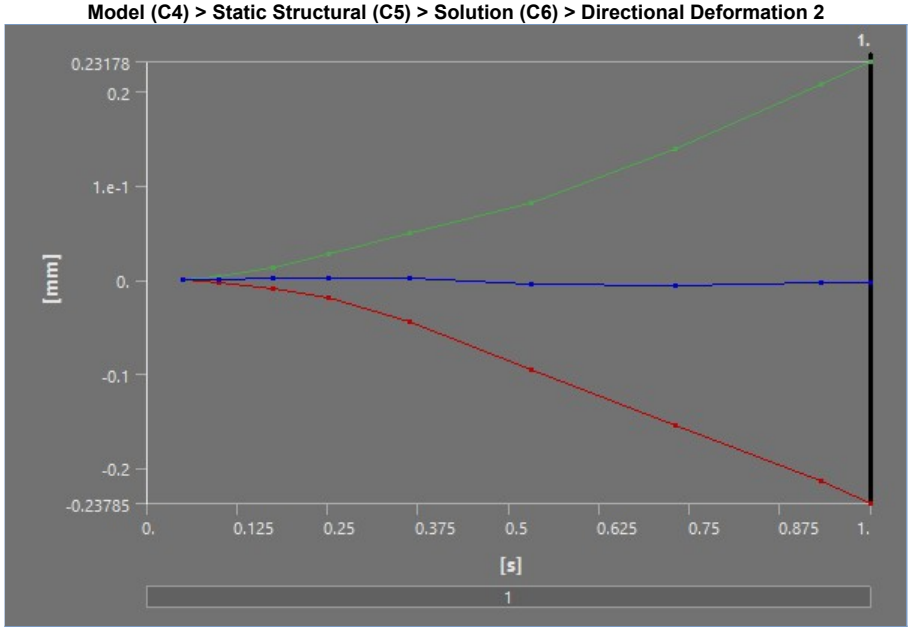



**TABLE 21**
**Model (C4) > Static Structural (C5) > Solution (C6) > Directional Deformation 2**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | -9.77e-015 | 6.6613e-015 | -3.2949e-016 |
| 0.1 | -2.6623e-003 | 2.8272e-003 | 4.1433e-005 |
| 0.175 | -9.1168e-003 | 1.2767e-002 | 1.0761e-003 |
| 0.25 | -1.9394e-002 | 2.7476e-002 | 2.3801e-003 |
| 0.3625 | -4.3938e-002 | 4.9343e-002 | 1.4006e-003 |
| 0.53125 | -9.5406e-002 | 8.1147e-002 | -4.6796e-003 |
| 0.73125 | -0.1552 | 0.13872 | -6.592e-003 |
| 0.93125 | -0.21314 | 0.20827 | -3.3871e-003 |
| 1. | -0.23785 | 0.23178 | -3.7474e-003 |

**FIGURE 7**
**Model (C4) > Static Structural (C5) > Solution (C6) > Directional Deformation 2 > Figure**

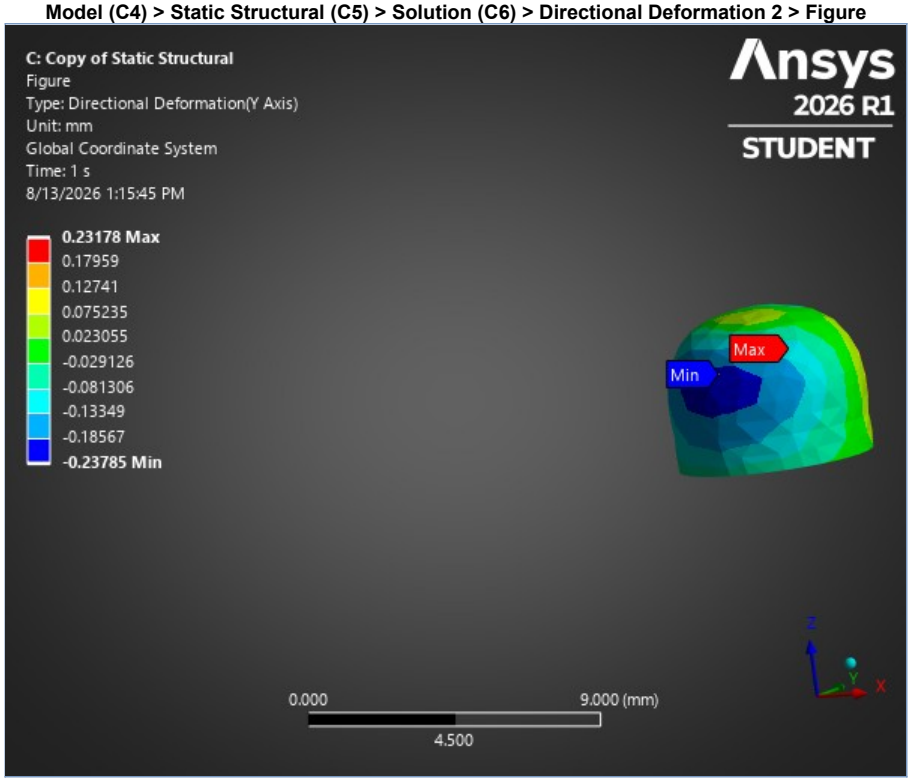


**FIGURE 8**
**Model (C4) > Static Structural (C5) > Solution (C6) > Directional Deformation 3**

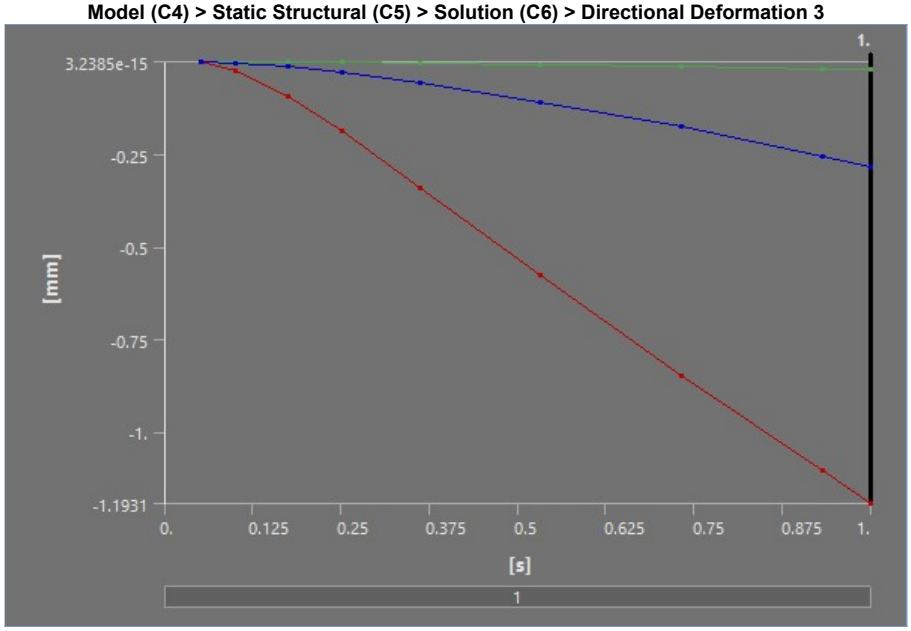



**TABLE 22**
**Model (C4) > Static Structural (C5) > Solution (C6) > Directional Deformation 3**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | -5.6811e-015 | 3.2385e-015 | -7.9462e-017 |
| 0.1 | -2.4527e-002 | -2.6168e-004 | -3.3642e-003 |
| 0.175 | -9.3166e-002 | -9.8988e-004 | -1.3402e-002 |
| 0.25 | -0.18562 | -2.0247e-003 | -2.8443e-002 |
| 0.3625 | -0.34029 | -4.2188e-003 | -5.6901e-002 |
| 0.53125 | -0.57795 | -8.3571e-003 | -0.10785 |
| 0.73125 | -0.8478 | -1.3208e-002 | -0.17638 |
| 0.93125 | -1.1054 | -1.9094e-002 | -0.25518 |
| 1. | -1.1931 | -2.1564e-002 | -0.28609 |

**FIGURE 9**
**Model (C4) > Static Structural (C5) > Solution (C6) > Directional Deformation 3 > Figure**

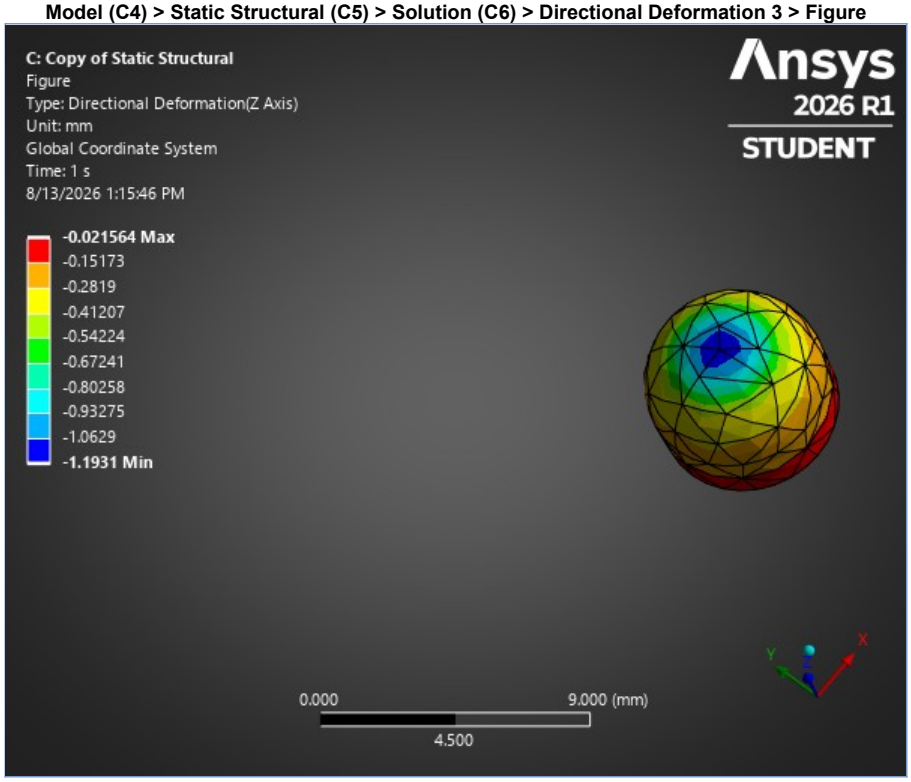


**FIGURE 10**
**Model (C4) > Static Structural (C5) > Solution (C6) > Equivalent Elastic Strain**

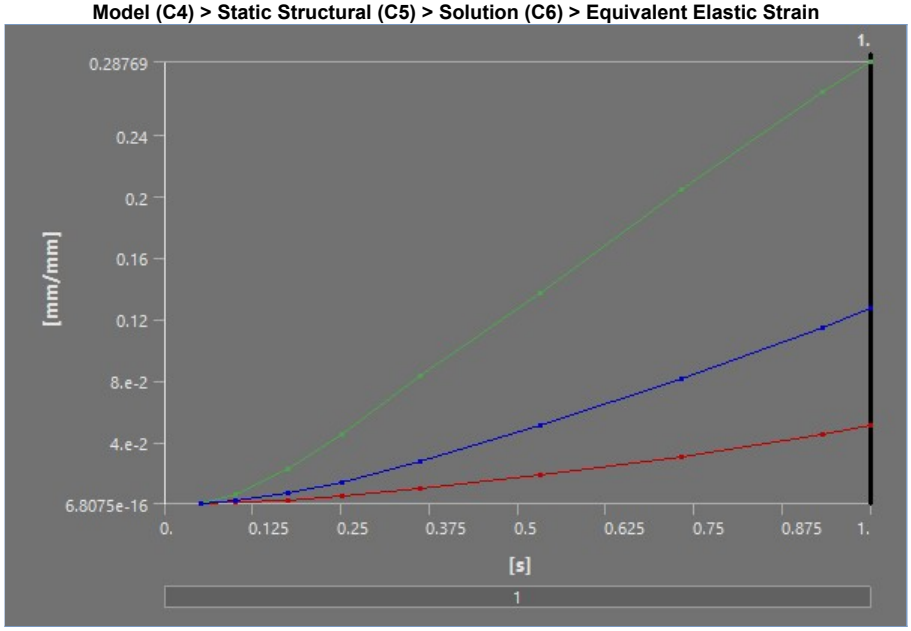



**TABLE 23**
**Model (C4) > Static Structural (C5) > Solution (C6) > Equivalent Elastic Strain**

| Time [s] | Minimum [mm/mm] | Maximum [mm/mm] | Average [mm/mm] |
|---|---|---|---|
| 5.e-002 | 6.8075e-016 | 5.4725e-015 | 1.4877e-015 |
| 0.1 | 5.8135e-004 | 6.1678e-003 | 1.6801e-003 |
| 0.175 | 2.3145e-003 | 2.2268e-002 | 6.6263e-003 |
| 0.25 | 4.8716e-003 | 4.5391e-002 | 1.3902e-002 |
| 0.3625 | 9.7923e-003 | 8.3135e-002 | 2.736e-002 |
| 0.53125 | 1.8664e-002 | 0.13652 | 5.0707e-002 |
| 0.73125 | 3.0555e-002 | 0.20426 | 8.125e-002 |
| 0.93125 | 4.4922e-002 | 0.26777 | 0.11484 |
| 1. | 5.0826e-002 | 0.28769 | 0.12745 |

**FIGURE 11**
**Model (C4) > Static Structural (C5) > Solution (C6) > Equivalent Elastic Strain > Figure**

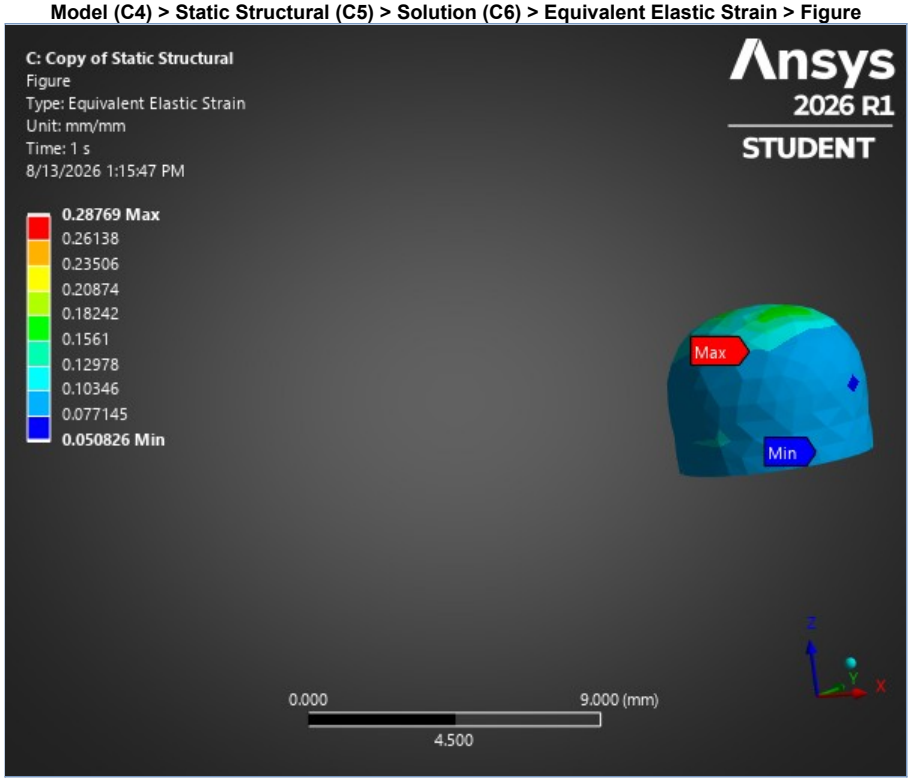


**FIGURE 12**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Principal Elastic Strain**

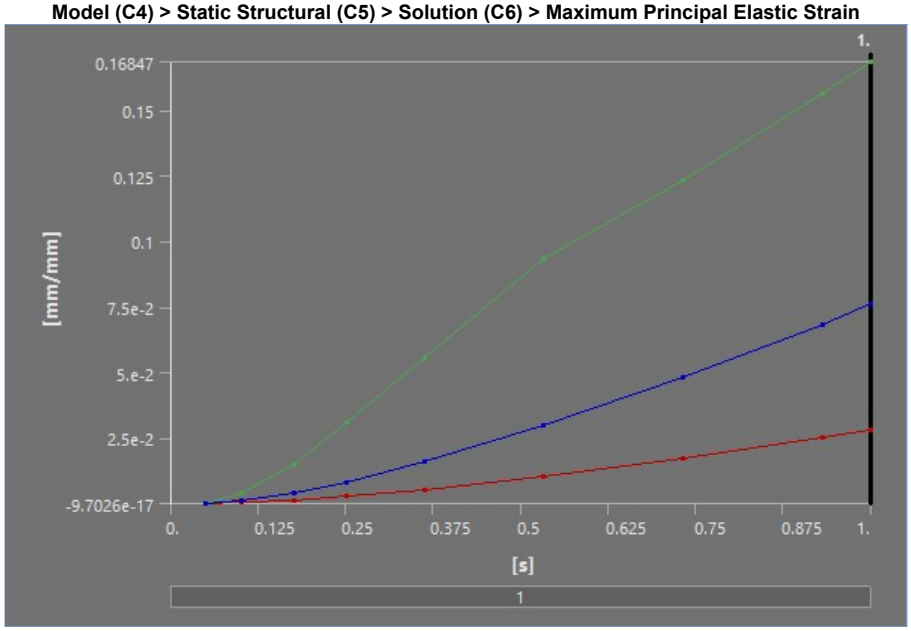



**TABLE 24**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Principal Elastic Strain**

| Time [s] | Minimum [mm/mm] | Maximum [mm/mm] | Average [mm/mm] |
|---|---|---|---|
| 5.e-002 | -9.7026e-017 | 2.8866e-015 | 3.7756e-016 |
| 0.1 | 3.249e-004 | 4.0603e-003 | 9.5944e-004 |
| 0.175 | 1.2984e-003 | 1.5164e-002 | 3.799e-003 |
| 0.25 | 2.7146e-003 | 3.0888e-002 | 8.0039e-003 |
| 0.3625 | 5.4276e-003 | 5.5566e-002 | 1.5861e-002 |
| 0.53125 | 1.0384e-002 | 9.3279e-002 | 2.9675e-002 |
| 0.73125 | 1.696e-002 | 0.12335 | 4.7946e-002 |
| 0.93125 | 2.5033e-002 | 0.15643 | 6.8305e-002 |
| 1. | 2.8335e-002 | 0.16847 | 7.5989e-002 |

**FIGURE 13**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Principal Elastic Strain > Figure**

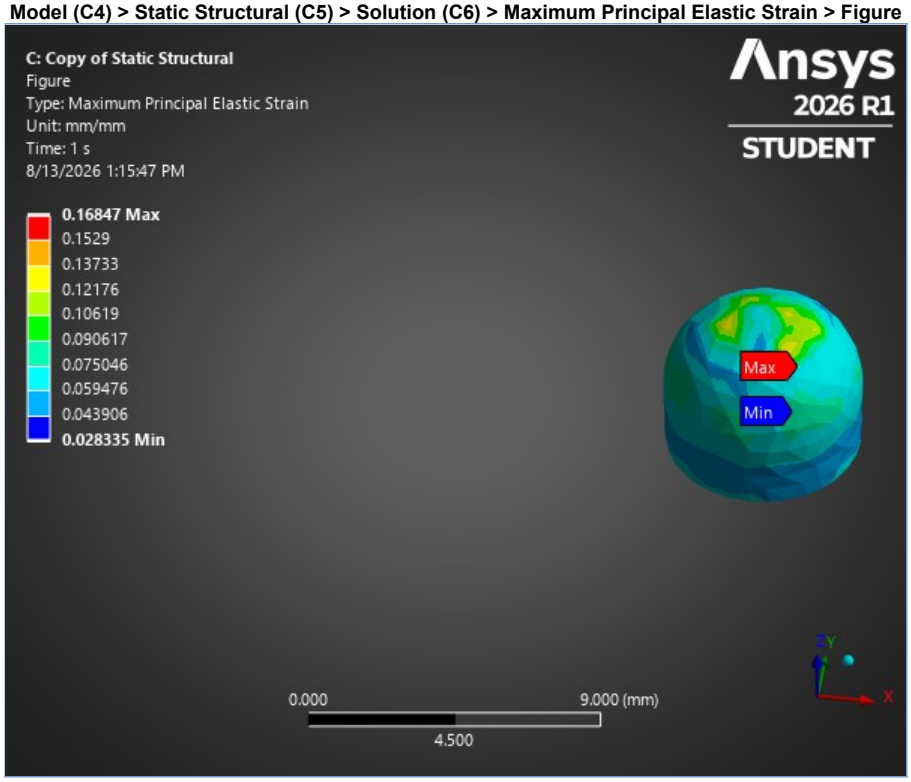


**FIGURE 14**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Shear Elastic Strain**

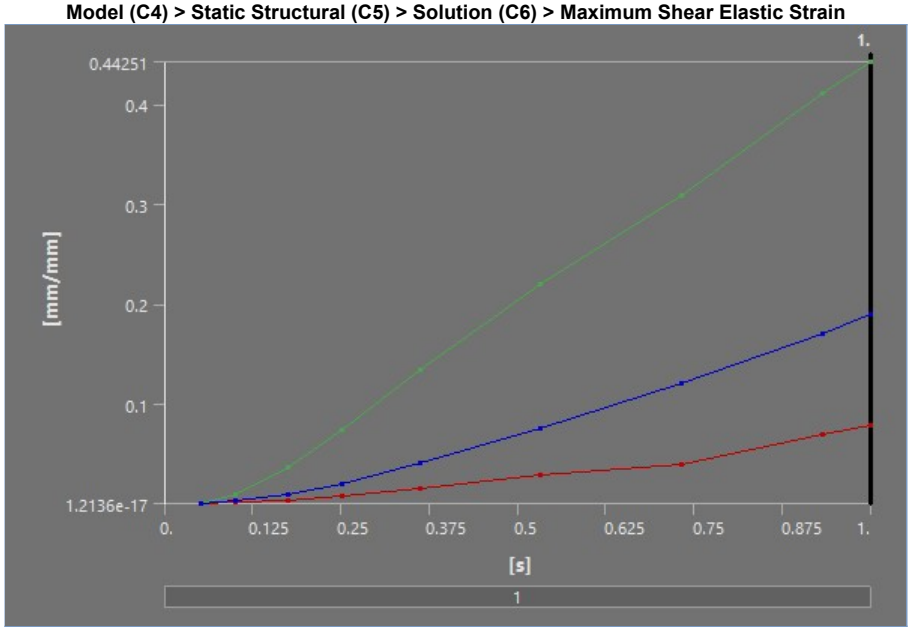



**TABLE 25**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Shear Elastic Strain**

| Time [s] | Minimum [mm/mm] | Maximum [mm/mm] | Average [mm/mm] |
|---|---|---|---|
| 5.e-002 | 1.2136e-017 | 6.9019e-015 | 8.0034e-016 |
| 0.1 | 8.8358e-004 | 9.276e-003 | 2.4368e-003 |
| 0.175 | 3.5271e-003 | 3.5882e-002 | 9.6323e-003 |
| 0.25 | 7.3999e-003 | 7.3087e-002 | 2.027e-002 |
| 0.3625 | 1.4836e-002 | 0.13328 | 4.0055e-002 |
| 0.53125 | 2.8572e-002 | 0.21991 | 7.4542e-002 |
| 0.73125 | 3.8742e-002 | 0.3085 | 0.1198 |
| 0.93125 | 6.8971e-002 | 0.41102 | 0.16996 |
| 1. | 7.7796e-002 | 0.44251 | 0.18891 |

**FIGURE 15**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Shear Elastic Strain > Figure**

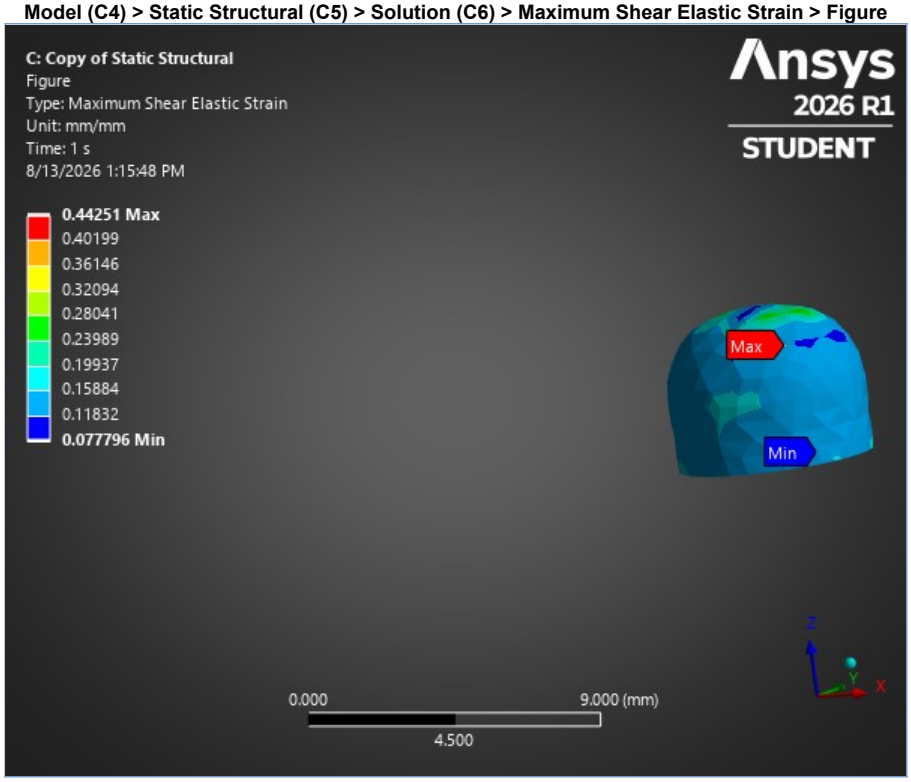


**FIGURE 16**
**Model (C4) > Static Structural (C5) > Solution (C6) > Elastic Strain Intensity**

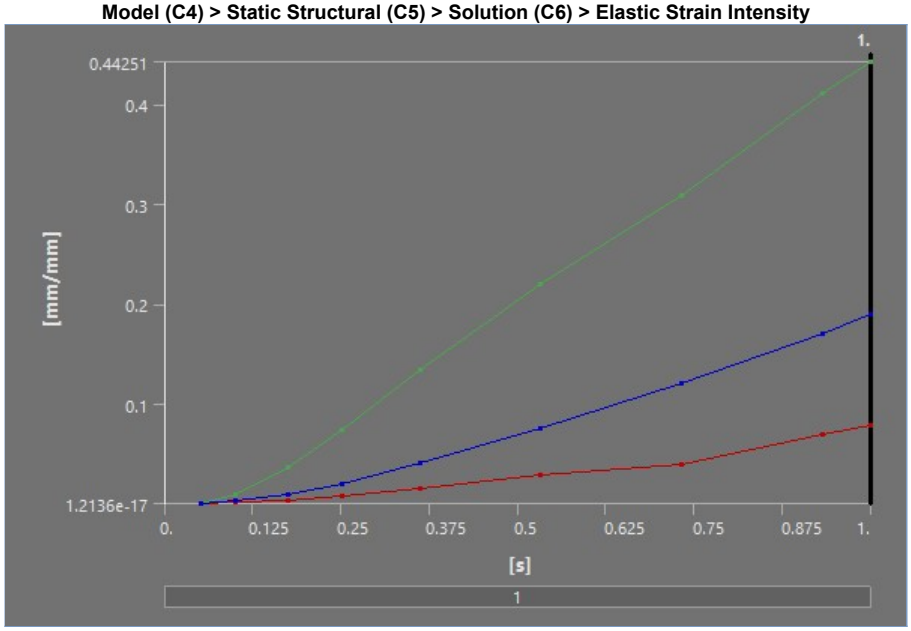



**TABLE 26**
**Model (C4) > Static Structural (C5) > Solution (C6) > Elastic Strain Intensity**

| Time [s] | Minimum [mm/mm] | Maximum [mm/mm] | Average [mm/mm] |
|---|---|---|---|
| 5.e-002 | 1.2136e-017 | 6.9019e-015 | 8.0034e-016 |
| 0.1 | 8.8358e-004 | 9.276e-003 | 2.4368e-003 |
| 0.175 | 3.5271e-003 | 3.5882e-002 | 9.6323e-003 |
| 0.25 | 7.3999e-003 | 7.3087e-002 | 2.027e-002 |
| 0.3625 | 1.4836e-002 | 0.13328 | 4.0055e-002 |
| 0.53125 | 2.8572e-002 | 0.21991 | 7.4542e-002 |
| 0.73125 | 3.8742e-002 | 0.3085 | 0.1198 |
| 0.93125 | 6.8971e-002 | 0.41102 | 0.16996 |
| 1. | 7.7796e-002 | 0.44251 | 0.18891 |

**FIGURE 17**
**Model (C4) > Static Structural (C5) > Solution (C6) > Elastic Strain Intensity > Figure**

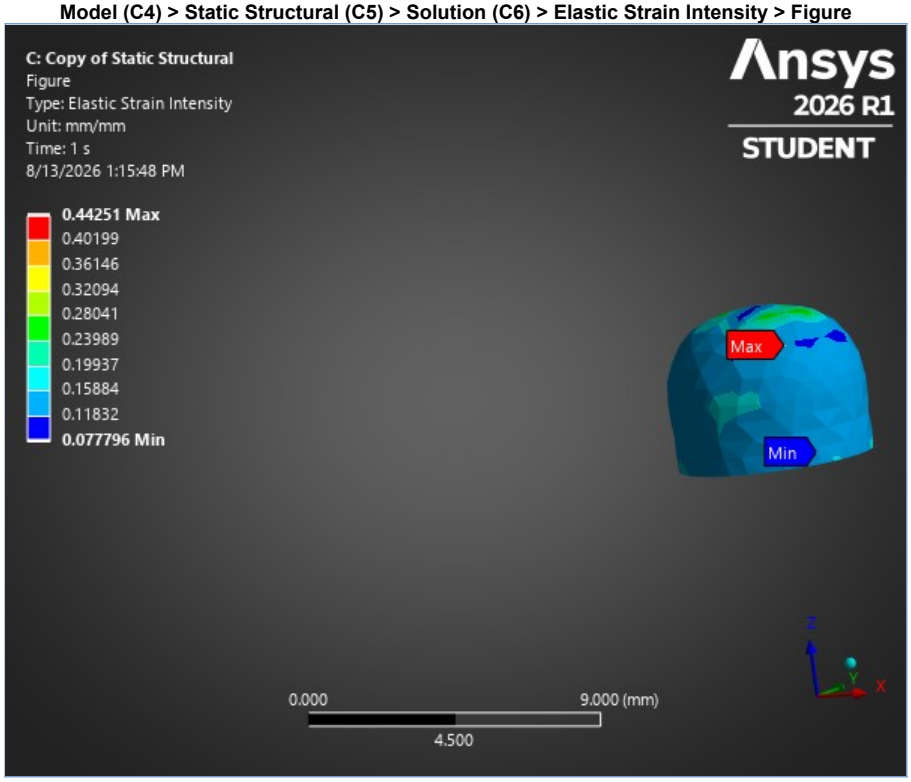


**FIGURE 18**
**Model (C4) > Static Structural (C5) > Solution (C6) > Equivalent Stress**

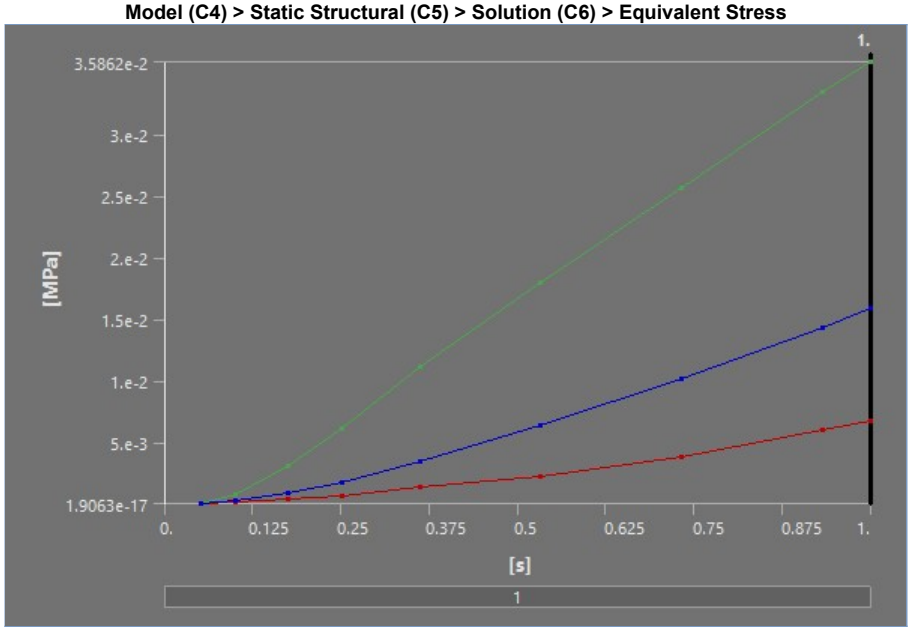



**TABLE 27**
**Model (C4) > Static Structural (C5) > Solution (C6) > Equivalent Stress**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | 1.9063e-017 | 9.4577e-016 | 1.2124e-016 |
| 0.1 | 7.9273e-005 | 7.87e-004 | 2.14e-004 |
| 0.175 | 3.148e-004 | 3.017e-003 | 8.4303e-004 |
| 0.25 | 6.634e-004 | 6.1014e-003 | 1.7667e-003 |
| 0.3625 | 1.3289e-003 | 1.1046e-002 | 3.4665e-003 |
| 0.53125 | 2.2335e-003 | 1.7929e-002 | 6.3828e-003 |
| 0.73125 | 3.7296e-003 | 2.5612e-002 | 1.0149e-002 |
| 0.93125 | 5.9937e-003 | 3.3435e-002 | 1.4269e-002 |
| 1. | 6.7548e-003 | 3.5862e-002 | 1.5816e-002 |

**FIGURE 19**
**Model (C4) > Static Structural (C5) > Solution (C6) > Equivalent Stress > Figure**

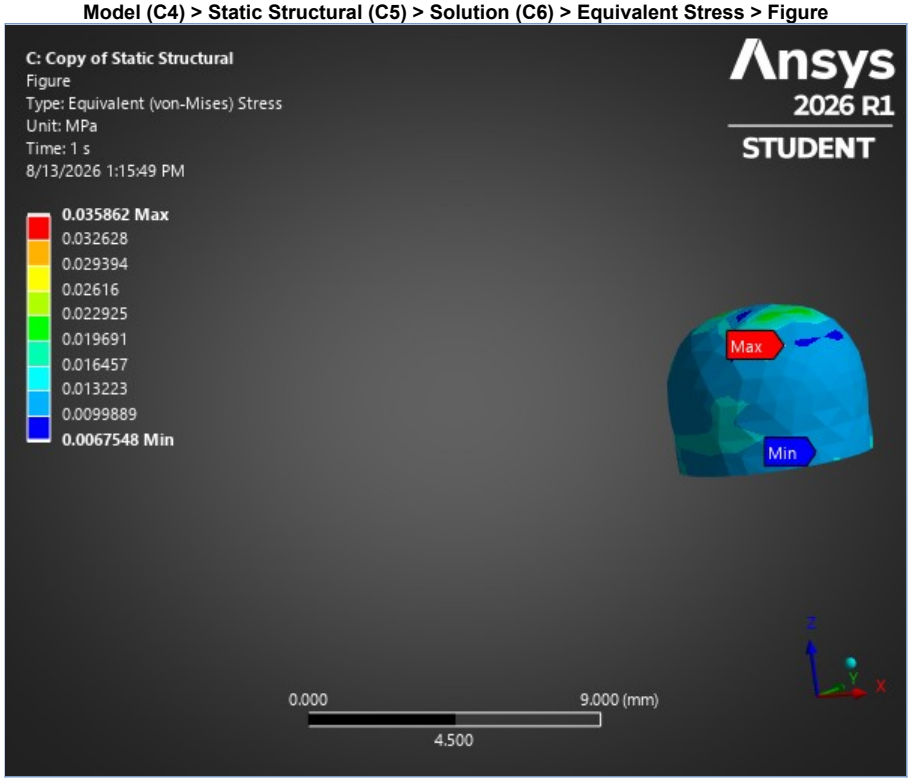


**FIGURE 20**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Principal Stress**

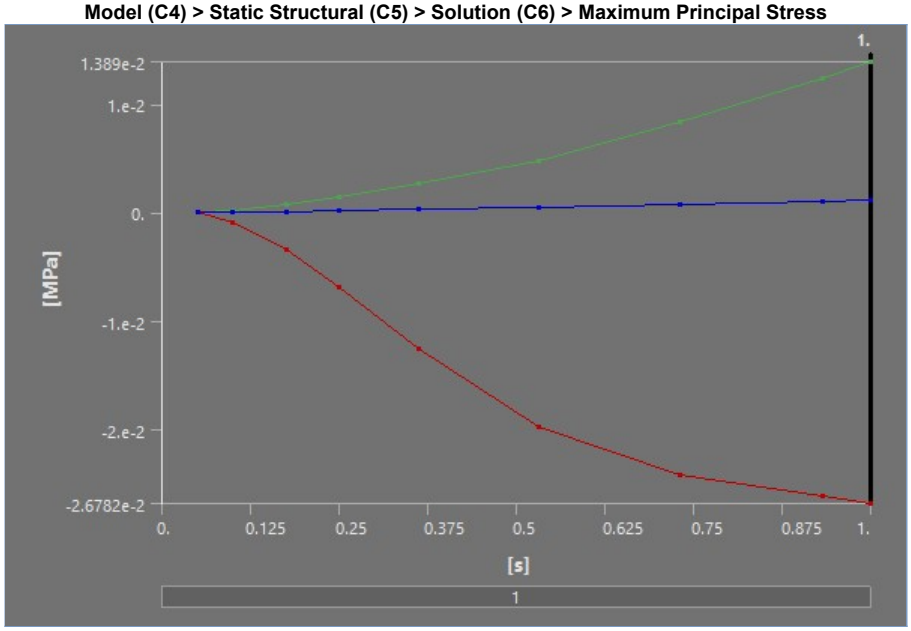



**TABLE 28**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Principal Stress**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | -4.2383e-015 | 7.0866e-015 | 3.3284e-016 |
| 0.1 | -8.918e-004 | 1.839e-004 | 1.8249e-005 |
| 0.175 | -3.363e-003 | 7.142e-004 | 7.069e-005 |
| 0.25 | -6.8236e-003 | 1.4327e-003 | 1.4364e-004 |
| 0.3625 | -1.2477e-002 | 2.7274e-003 | 2.7214e-004 |
| 0.53125 | -1.9673e-002 | 4.7954e-003 | 4.8672e-004 |
| 0.73125 | -2.4112e-002 | 8.298e-003 | 7.7423e-004 |
| 0.93125 | -2.6107e-002 | 1.2409e-002 | 1.0705e-003 |
| 1. | -2.6782e-002 | 1.389e-002 | 1.1424e-003 |

**FIGURE 21**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Principal Stress > Figure**

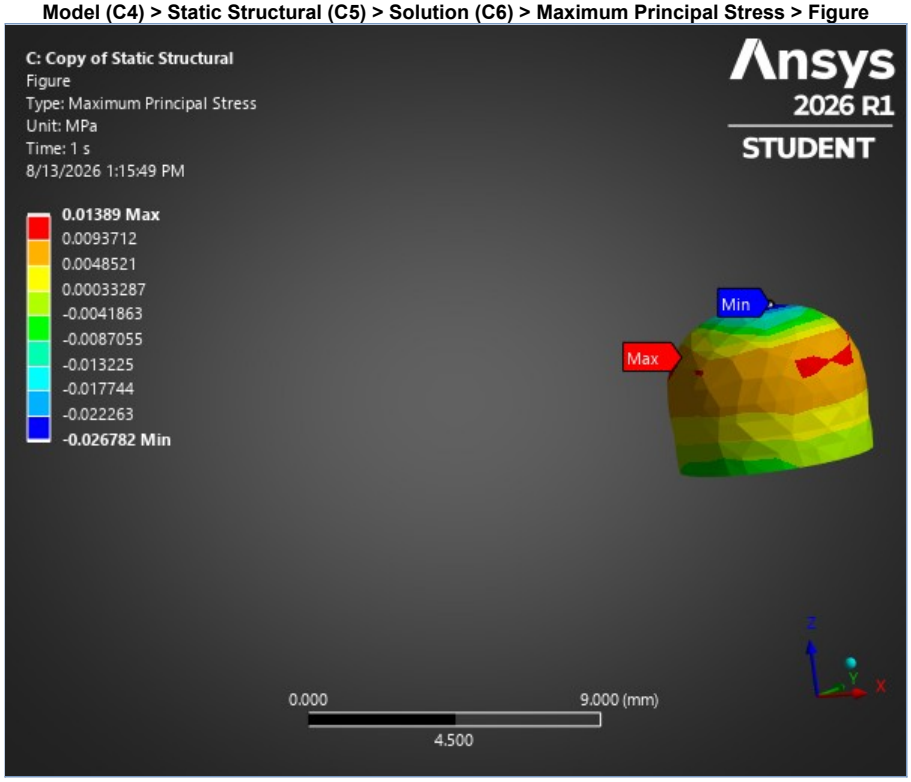


**FIGURE 22**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Shear Stress**

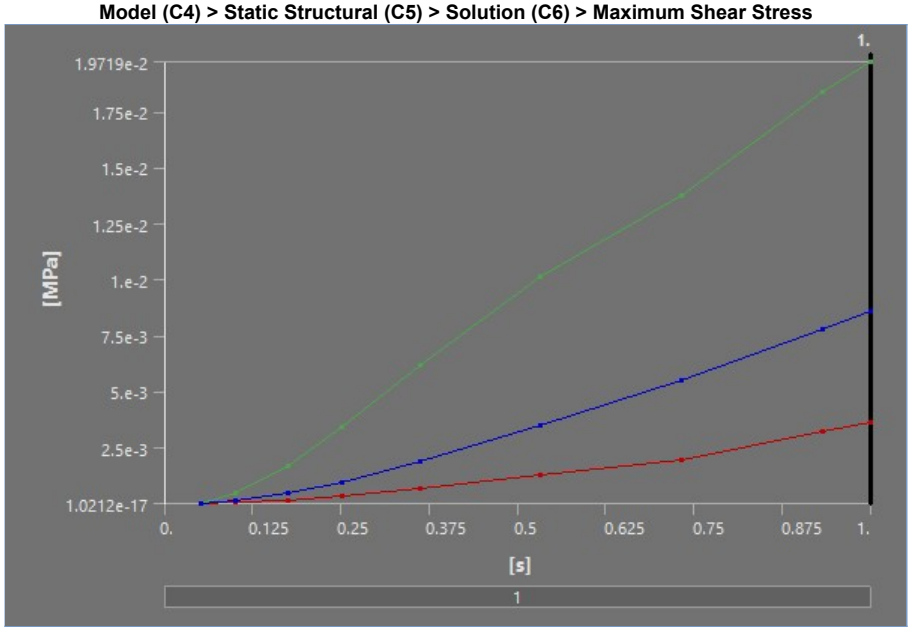



**TABLE 29**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Shear Stress**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | 1.0212e-017 | 5.4528e-016 | 6.8209e-017 |
| 0.1 | 4.1784e-005 | 4.3809e-004 | 1.1516e-004 |
| 0.175 | 1.6669e-004 | 1.6881e-003 | 4.5409e-004 |
| 0.25 | 3.4936e-004 | 3.4212e-003 | 9.5239e-004 |
| 0.3625 | 6.991e-004 | 6.184e-003 | 1.8717e-003 |
| 0.53125 | 1.2553e-003 | 1.0142e-002 | 3.4564e-003 |
| 0.73125 | 1.9753e-003 | 1.3747e-002 | 5.5089e-003 |
| 0.93125 | 3.2131e-003 | 1.8348e-002 | 7.7588e-003 |
| 1. | 3.6168e-003 | 1.9719e-002 | 8.6032e-003 |

**FIGURE 23**
**Model (C4) > Static Structural (C5) > Solution (C6) > Maximum Shear Stress > Figure**

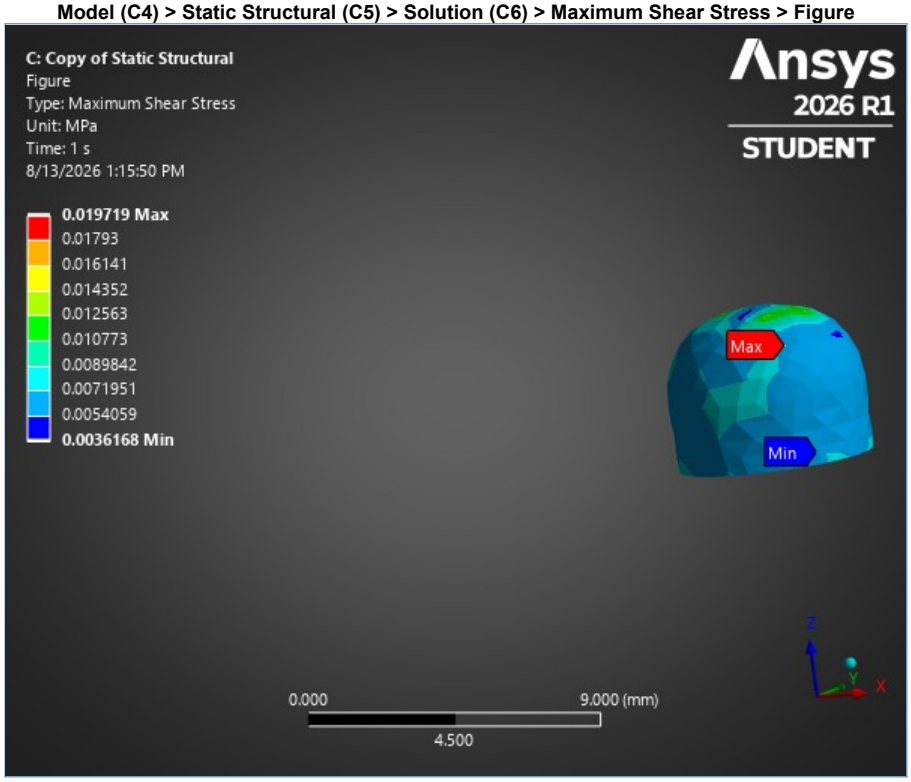


**TABLE 30**
**Model (C4) > Static Structural (C5) > Solution (C6) > Results**

| Object Name | *Stress Intensity* | *Strain Energy* |
|---|---|---|
| State | Solved | |
| **Scope** | | |
| Scoping Method | Geometry Selection | |
| Geometry | 1 Body | |
| **Definition** | | |
| Type | Stress Intensity | Strain Energy |
| By | Time | |
| Display Time | Last | |
| Separate Data by Entity | No | |
| Calculate Time History | Yes | |
| Identifier | | |
| Suppressed | No | |
| **Integration Point Results** | | |
| Display Option | Averaged | |
| Average Across Bodies | No | |
| **Results** | | |
| Minimum | 7.2336e-003 MPa | 3.3012e-005 mJ |
| Maximum | 3.9438e-002 MPa | 2.962e-003 mJ |
| Average | 1.7206e-002 MPa | |
| Minimum Occurs On | RUN 6 (6MM)\|Body9 | |
| Maximum Occurs On | RUN 6 (6MM)\|Body9 | |
| Total | | 0.13873 mJ |
| **Minimum Value Over Time** | | |
| Minimum | 2.0424e-017 MPa | -8.3732e-018 mJ |
| Maximum | 7.2336e-003 MPa | 3.3012e-005 mJ |
| **Maximum Value Over Time** | | |
| Minimum | 1.0906e-015 MPa | 3.5245e-018 mJ |
| Maximum | 3.9438e-002 MPa | 2.962e-003 mJ |
| **Information** | | |
| Time | 1. s | |
| Load Step | 1 | |
| Substep | 9 | |
| Iteration Number | 30 | |

**FIGURE 24**
**Model (C4) > Static Structural (C5) > Solution (C6) > Stress Intensity**

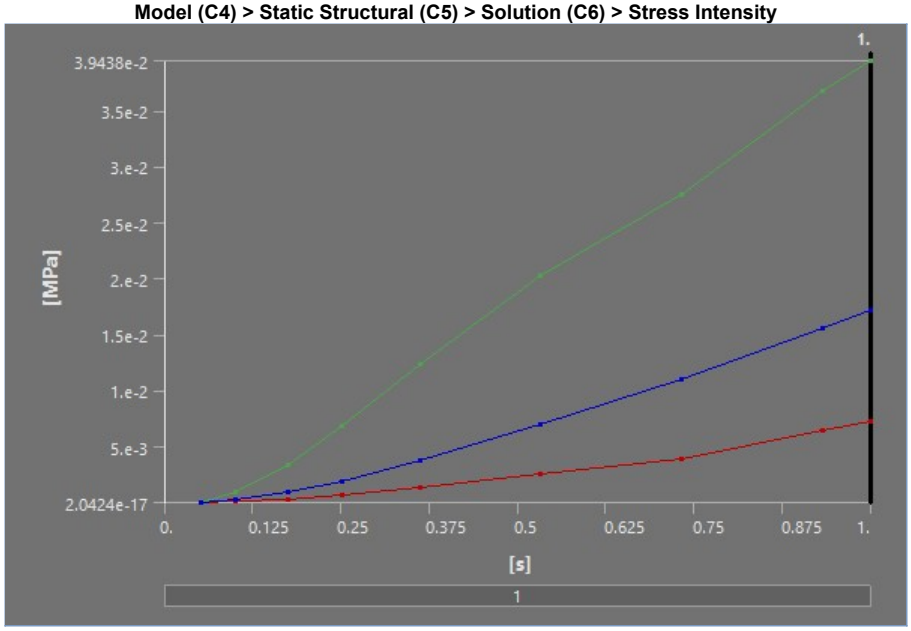



**TABLE 31**
**Model (C4) > Static Structural (C5) > Solution (C6) > Stress Intensity**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | 2.0424e-017 | 1.0906e-015 | 1.3642e-016 |
| 0.1 | 8.3568e-005 | 8.7618e-004 | 2.3032e-004 |
| 0.175 | 3.3338e-004 | 3.3763e-003 | 9.0818e-004 |
| 0.25 | 6.9871e-004 | 6.8423e-003 | 1.9048e-003 |

| | | | |
|---|---|---|---|
| 0.3625 | 1.3982e-003 | 1.2368e-002 | 3.7435e-003 |
| 0.53125 | 2.5105e-003 | 2.0284e-002 | 6.9128e-003 |
| 0.73125 | 3.9506e-003 | 2.7495e-002 | 1.1018e-002 |
| 0.93125 | 6.4263e-003 | 3.6696e-002 | 1.5518e-002 |
| 1. | 7.2336e-003 | 3.9438e-002 | 1.7206e-002 |

**FIGURE 25**
**Model (C4) > Static Structural (C5) > Solution (C6) > Stress Intensity > Figure**

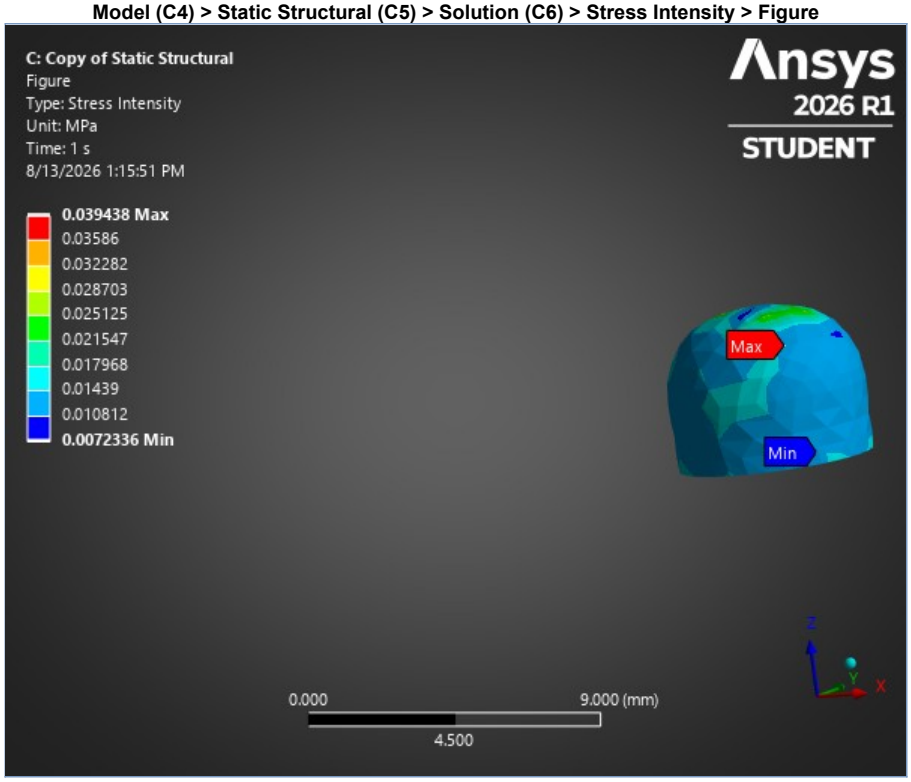



**FIGURE 26**
**Model (C4) > Static Structural (C5) > Solution (C6) > Strain Energy**

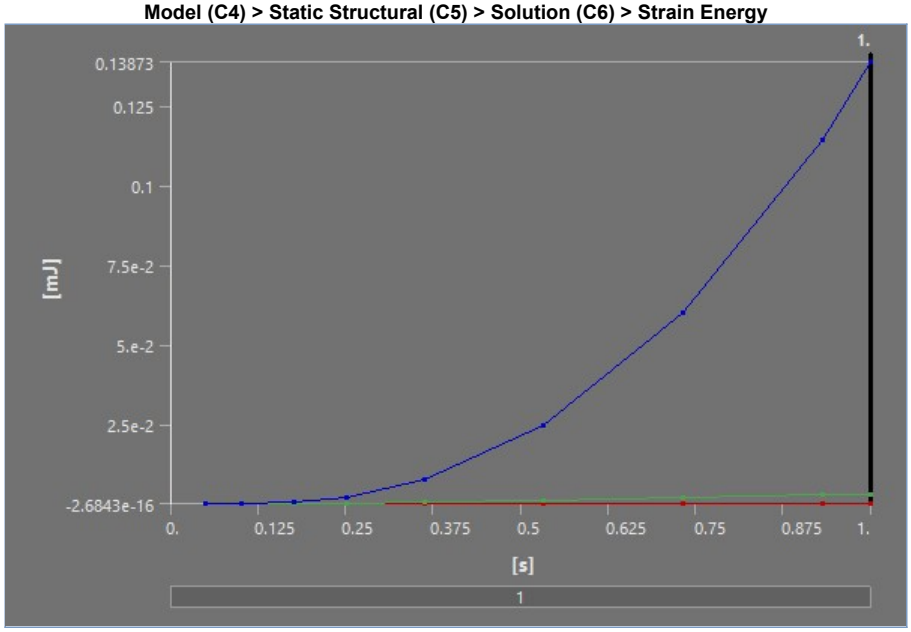



**TABLE 32**
**Model (C4) > Static Structural (C5) > Solution (C6) > Strain Energy**

| Time [s] | Minimum [mJ] | Maximum [mJ] | Total [mJ] |
|---|---|---|---|
| 5.e-002 | -8.3732e-018 | 3.5245e-018 | -2.6843e-016 |
| 0.1 | 3.9753e-009 | 1.9389e-006 | 3.3619e-005 |
| 0.175 | 6.2703e-008 | 2.6902e-005 | 4.9822e-004 |

| | | | |
|---|---|---|---|
| 0.25 | 2.7802e-007 | 1.0195e-004 | 2.0842e-003 |
| 0.3625 | 1.1319e-006 | 3.2316e-004 | 7.6233e-003 |
| 0.53125 | 4.3022e-006 | 8.7001e-004 | 2.4535e-002 |
| 0.73125 | 1.1801e-005 | 1.7433e-003 | 5.9744e-002 |
| 0.93125 | 2.5604e-005 | 2.6726e-003 | 0.11436 |
| 1. | 3.3012e-005 | 2.962e-003 | 0.13873 |

**FIGURE 27**
**Model (C4) > Static Structural (C5) > Solution (C6) > Strain Energy > Figure**

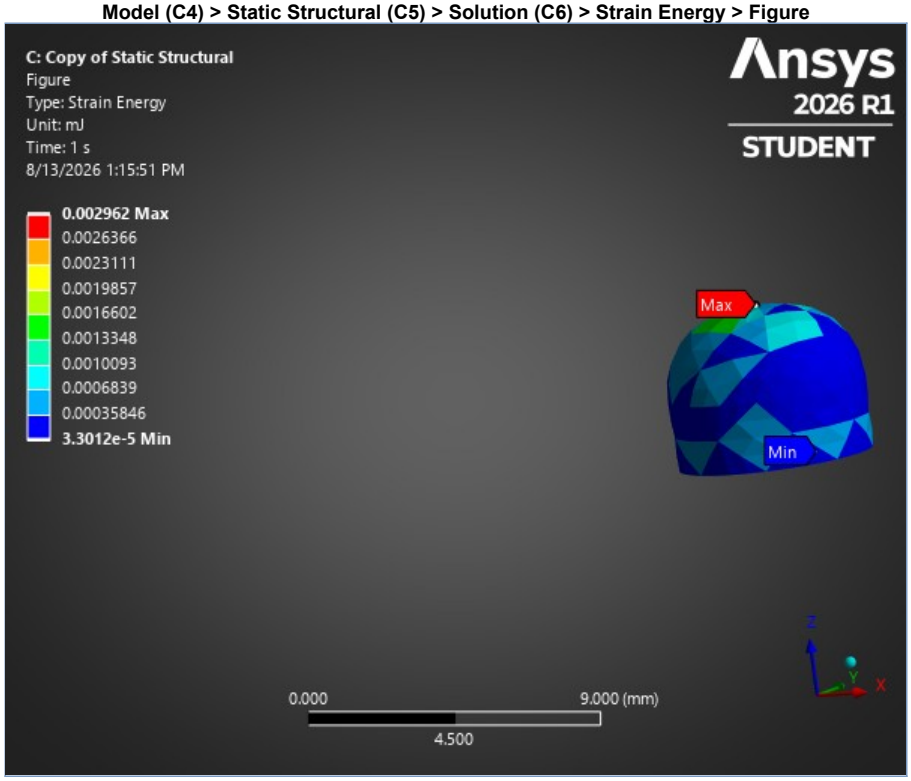



**TABLE 33**
**Model (C4) > Static Structural (C5) > Solution (C6) > Contact Tools**

| Object Name | *Contact Tool* |
|---|---|
| State | Solved |
| **Scope** | |
| Scoping Method | Worksheet |

**Model (C4) > Static Structural (C5) > Solution (C6) > Contact Tool**

| Name | Contact Side |
|---|---|
| Contact Region | Both |
| Contact Region 2 | Both |
| Contact Region 3 | Both |
| Contact Region 4 | Both |
| Contact Region 5 | Both |
| Contact Region 6 | Both |
| Contact Region 7 | Both |
| Contact Region 8 | Both |
| Contact Region 9 | Both |
| Frictionless - RUN 6 (6MM)\|Body9 To RUN 6 (6MM)\|Body12 | Both |

**TABLE 34**
**Model (C4) > Static Structural (C5) > Solution (C6) > Contact Tool > Results**

| Object Name | *Status* |
|---|---|
| State | Solved |
| **Definition** | |
| Type | Status |
| By | Time |
| Display Time | Last |
| Calculate Time History | Yes |
| Identifier | |
| Suppressed | No |

| Integration Point Results | |
|---|---|
| Display Option | Averaged |
| **Information** | |
| Time | 1. s |
| Load Step | 1 |
| Substep | 9 |
| Iteration Number | 30 |

**FIGURE 28**
**Model (C4) > Static Structural (C5) > Solution (C6) > Contact Tool > Status**

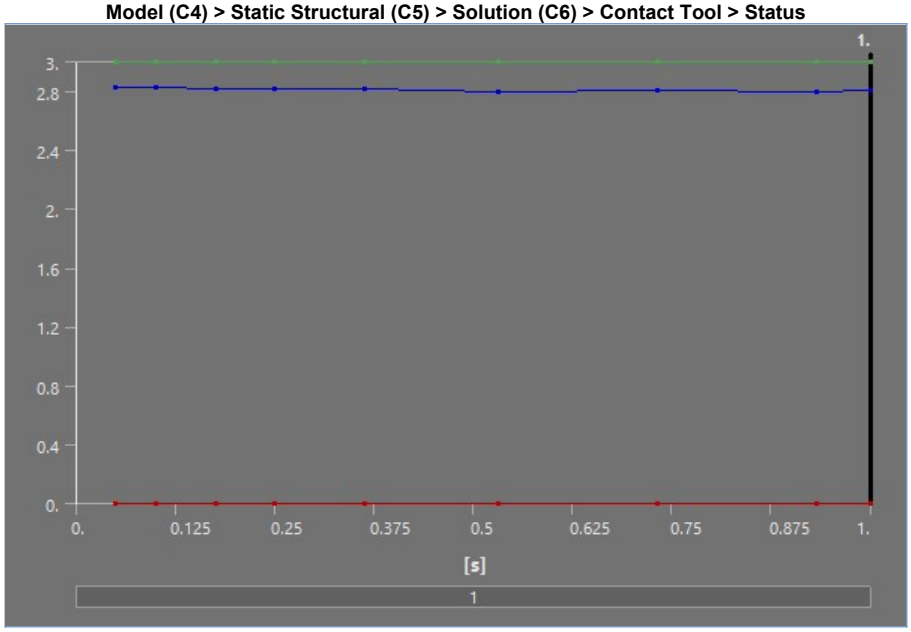



**TABLE 35**
**Model (C4) > Static Structural (C5) > Solution (C6) > Contact Tool > Status**

| Time [s] | Minimum | Maximum | Average |
|---|---|---|---|
| 5.e-002 | 0. | 3. | 2.8251 |
| 0.1 | | | 2.8293 |
| 0.175 | | | 2.8197 |
| 0.25 | | | 2.8156 |
| 0.3625 | | | 2.8153 |
| 0.53125 | | | 2.7942 |
| 0.73125 | | | 2.8071 |
| 0.93125 | | | 2.7995 |
| 1. | | | 2.8045 |

**FIGURE 29**
**Model (C4) > Static Structural (C5) > Solution (C6) > Contact Tool > Figure**

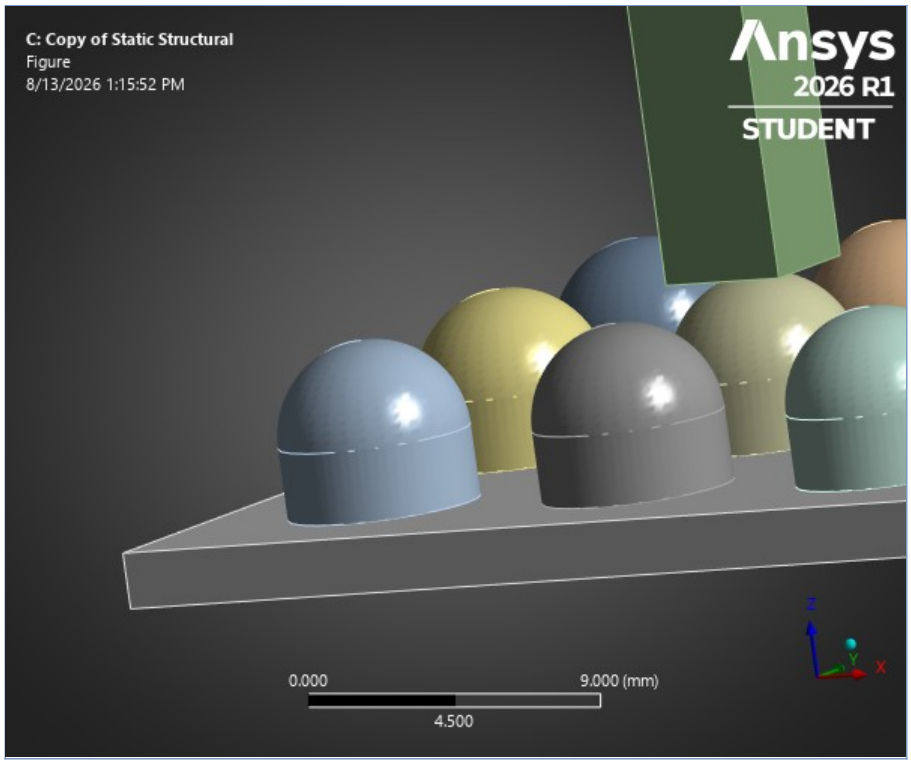



**TABLE 36**
**Model (C4) > Static Structural (C5) > Solution (C6) > Probes**

| Object Name | *Force Reaction* | *Moment Reaction* |
|---|---|---|
| State | Solved | |
| **Definition** | | |
| Type | Force Reaction | Moment Reaction |
| Location Method | Boundary Condition | |
| Boundary Condition | Displacement | |
| Orientation | Global Coordinate System | |
| Suppressed | No | |
| Summation | | Centroid |
| **Options** | | |
| Result Selection | All | |
| Display Time | End Time | |
| **Results** | | |
| X Axis | -1.3648e-003 N | -3.6283e-003 N·mm |
| Y Axis | -7.205e-004 N | -2.6296e-003 N·mm |
| Z Axis | -0.32662 N | 2.3646e-004 N·mm |
| Total | 0.32662 N | 4.4873e-003 N·mm |
| **Maximum Value Over Time** | | |
| X Axis | 4.7799e-004 N | 1.8498e-003 N·mm |
| Y Axis | 5.8795e-004 N | 2.6807e-004 N·mm |
| Z Axis | -2.5495e-006 N | 3.5893e-004 N·mm |
| Total | 0.32662 N | 7.8522e-003 N·mm |
| **Minimum Value Over Time** | | |
| X Axis | -1.4085e-003 N | -6.0167e-003 N·mm |
| Y Axis | -1.6005e-003 N | -5.045e-003 N·mm |
| Z Axis | -0.32662 N | -5.8628e-005 N·mm |
| Total | 2.5502e-006 N | 1.1087e-006 N·mm |
| **Information** | | |
| Time | 1. s | |
| Load Step | 1 | |
| Substep | 9 | |
| Iteration Number | 30 | |

**FIGURE 30**
**Model (C4) > Static Structural (C5) > Solution (C6) > Force Reaction**

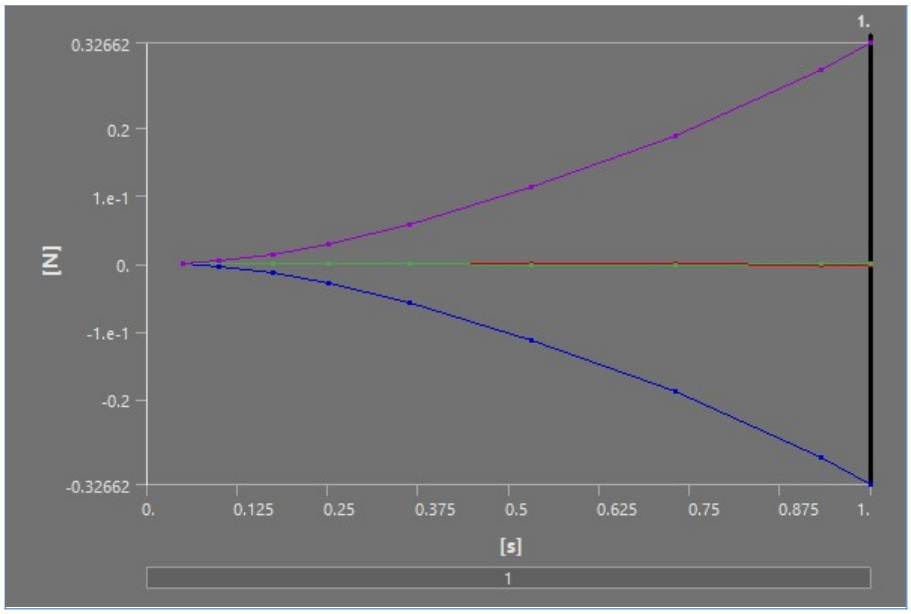



**TABLE 37**
**Model (C4) > Static Structural (C5) > Solution (C6) > Force Reaction**

| Time [s] | Force Reaction (X) [N] | Force Reaction (Y) [N] | Force Reaction (Z) [N] | Force Reaction (Total) [N] |
|---|---|---|---|---|
| 5.e-002 | -4.1398e-008 | -4.1398e-008 | -2.5495e-006 | 2.5502e-006 |
| 0.1 | 5.2399e-005 | 1.4284e-005 | -3.4102e-003 | 3.4106e-003 |
| 0.175 | -6.8954e-005 | 2.7051e-004 | -1.3624e-002 | 1.3627e-002 |
| 0.25 | 3.0535e-005 | 5.8795e-004 | -2.9064e-002 | 2.907e-002 |
| 0.3625 | 2.2756e-004 | 2.9123e-004 | -5.8864e-002 | 5.8866e-002 |
| 0.53125 | 4.7799e-004 | -1.1667e-003 | -0.11302 | 0.11303 |
| 0.73125 | 3.9874e-004 | -1.6005e-003 | -0.18994 | 0.18995 |
| 0.93125 | -1.4085e-003 | -7.3016e-004 | -0.28671 | 0.28672 |
| 1. | -1.3648e-003 | -7.205e-004 | -0.32662 | 0.32662 |

**FIGURE 31**
**Model (C4) > Static Structural (C5) > Solution (C6) > Force Reaction > Figure**

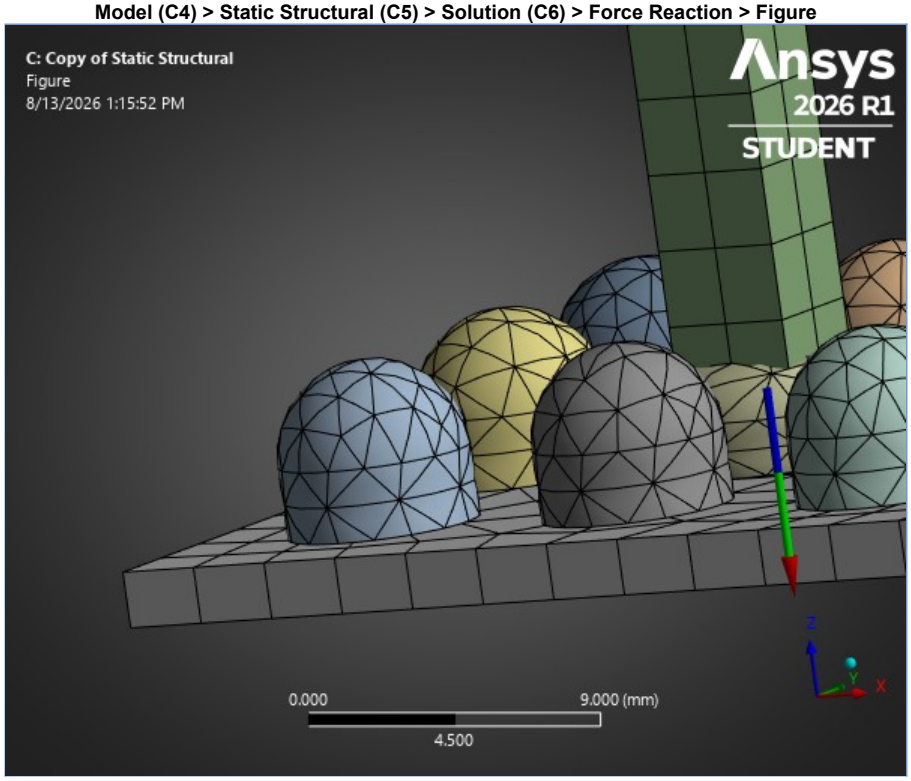



**FIGURE 32**

**Model (C4) > Static Structural (C5) > Solution (C6) > Moment Reaction**

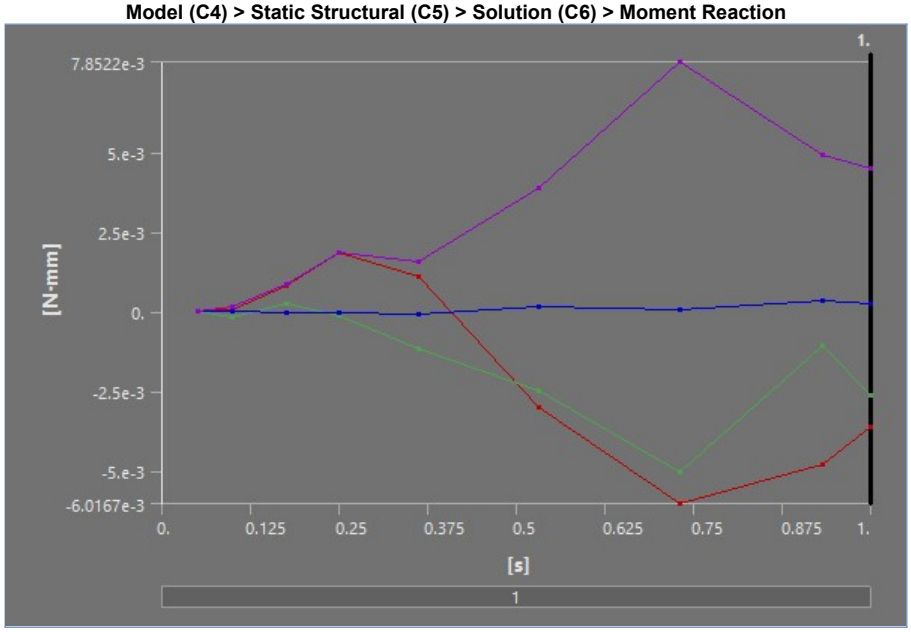
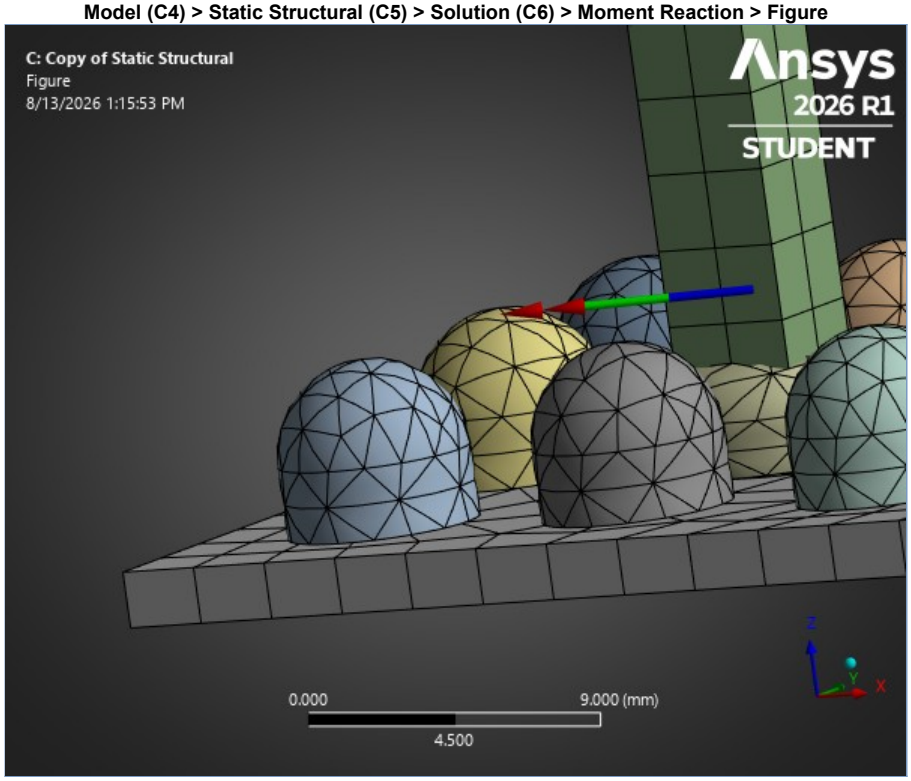



**TABLE 38**
**Model (C4) > Static Structural (C5) > Solution (C6) > Moment Reaction**

| Time [s] | Moment Reaction (X) [N·mm] | Moment Reaction (Y) [N·mm] | Moment Reaction (Z) [N·mm] | Moment Reaction (Total) [N·mm] |
|---|---|---|---|---|
| 5.e-002 | 7.8398e-007 | -7.8398e-007 | 7.6245e-014 | 1.1087e-006 |
| 0.1 | 4.8055e-005 | -1.4913e-004 | 6.5992e-007 | 1.5668e-004 |
| 0.175 | 8.4042e-004 | 2.6807e-004 | -7.8399e-006 | 8.8217e-004 |
| 0.25 | 1.8498e-003 | -1.0285e-004 | -2.1656e-005 | 1.8528e-003 |
| 0.3625 | 1.1025e-003 | -1.1404e-003 | -5.8628e-005 | 1.5873e-003 |
| 0.53125 | -2.9835e-003 | -2.5e-003 | 1.5474e-004 | 3.8955e-003 |
| 0.73125 | -6.0167e-003 | -5.045e-003 | 6.765e-005 | 7.8522e-003 |
| 0.93125 | -4.8033e-003 | -1.0832e-003 | 3.5893e-004 | 4.937e-003 |
| 1. | -3.6283e-003 | -2.6296e-003 | 2.3646e-004 | 4.4873e-003 |

**FIGURE 33**
**Model (C4) > Static Structural (C5) > Solution (C6) > Moment Reaction > Figure**



# Material Data

## *ecoflex 00-45*

**TABLE 39**
**ecoflex 00-45 > Constants**

| | |
|---|---|
| Density | 1.06e-006 kg mm^-3 |

**TABLE 40**
**ecoflex 00-45 > Color**

| Red | Green | Blue |
|---|---|---|
| 181 | 194 | 156 |

**TABLE 41**
**ecoflex 00-45 > Neo-Hookean**

| Initial Shear Modulus Mu MPa | Incompressibility Parameter D1 MPa^-1 | Temperature K |
|---|---|---|
| 4.73e-002 | 0.424 | |

## *Plastic, ABS (high-impact)*

**TABLE 42**
**Plastic, ABS (high-impact) > Constants**

| | |
|---|---|
| Density | 1.03e-006 kg mm^-3 |
| Tensile Yield Strength | 43.82 MPa |
| Tensile Ultimate Strength | 36.26 MPa |
| Coefficient of Thermal Expansion | 1.84e-004 K^-1 |
| Thermal Conductivity | 1.997e-004 W mm^-1 K^-1 |
| Specific Heat | 1.4e+006 mJ kg^-1 K^-1 |
| Resistivity | 9.95e+016 ohm mm |
| Electric Loss Tangent | 5.916e-003 |
| Relative Permittivity | 2.993 |

**TABLE 43**
**Plastic, ABS (high-impact) > Opacity**

| Red | Green | Blue |
|---|---|---|
| 0 | 153 | 255 |

| Opacity |
|---|
| 0.9 |

| Metallic Finish |
|---|
| 0 |

**TABLE 44**
**Plastic, ABS (high-impact) > Isotropic Elasticity**

| Young's Modulus MPa | Poisson's Ratio | Bulk Modulus MPa | Shear Modulus MPa | Temperature K |
|---|---|---|---|---|
| 2405 | 0.4089 | 4399.9 | 853.5 | 296.15 |

**TABLE 45**
**Plastic, ABS (high-impact) > Isotropic Secant Coefficient of Thermal Expansion**

| Zero-Thermal-Strain Reference Temperature K |
|---|
| 296.15 |

**TABLE 46**
**Plastic, ABS (high-impact) > Climate change (CO2-eq)**

| Climate change (CO2-eq) dBA |
|---|
| 3.661 |

**TABLE 47**
**Plastic, ABS (high-impact) > Embodied energy**

| Embodied energy mJ kg^-1 |
|---|
| 9.64e+010 |

**TABLE 48**
**Plastic, ABS (high-impact) > Recycle**

| Recycle |
|---|
| 1 |